

Configuration Homology: A Geometric Approach To Homology

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Contents

1	Introduction	2
1.1	Conventions and Prerequisites	3
2	Configuration Homology	4
2.1	Constructing an Isomorphism From Singular Homology to Configuration Homology	5
2.2	Decomposable Maps	7
2.3	Ensembles	16
2.4	Geometric Realizations of Simplicial Sets are Hyperlocally Contractible	20
3	The Dold-Thom Theorem	22
3.1	The Grothendieck Completion Map	23
3.2	The Generalized Dold-Thom Theorem	26
3.3	The Hurewicz Map	29
4	Stable Homotopy	30
4.1	Stability of Configuration Homology	30
4.2	Spectra Induce Homology Theories	34
4.3	Spectra Induce Cohomology Theories	36
5	Configuration Cohomology	39
5.1	Representing Basic Cohomology Operations	41
5.2	The Cup Product	42
5.3	A Representation of Sq^1	43
6	Conclusion	48

1 Introduction

In this thesis, we will construct a geometric link between homology groups and homotopy groups, allowing us to describe the former in terms of the latter. For any abelian group G and topological space X , we will construct a topological abelian group $G^{(X)}$ such that $H_n(X; G) \cong \pi_n(G^{(X)})$. This space $G^{(X)}$ will have as elements formal sums of points in X with coefficients in G , where continuous movement through $G^{(X)}$ is to be thought of as continuously moving summands in a formal sum. The benefit of this is that it will give us a geometric way to think about homology groups, as well as help establish other connections between homology groups and homotopy groups. It will also give ground to the intuition that homology groups are an “abelianization” of homotopy groups. We will refer to the homology theory H_*^{conf} given by $X \mapsto \pi_*(G^{(X)})$ as configuration homology.

Configuration Homology can be seen as a generalization of the Dold-Thom Theorem. The Dold-Thom Theorem states that on based connected CW-complexes (X, b) , the functors $X \mapsto \pi_n(\text{Sp}(X))$ form a reduced homology theory, where $\text{Sp}(X)$ is a space whose elements are finite unordered tuples in X [3]. We explore this connection in more detail in Section 3.

In Section 2, we will give a mostly elementary and entirely original proof showing that H_*^{conf} is a homology theory. In the process, to deal with the pathological nature of the space $G^{(X)}$, we will develop a tool which we will call decomposition structures on topological groups. This tool will be general enough that we will be able to use it to study topological groups outside the setting of configuration homology. Notably, we will use it in Section 3 to prove that for certain topological monoids, their Grothendieck completion maps are weak equivalences.

We will also explore how configuration homology interacts with other homology theories like singular homology, the homology theory given by the Dold-Thom theorem, and homology theories represented by certain spectra. The last of these will allow us to give a natural dual to configuration homology: configuration cohomology. In essence, we will show that the spaces $G_0^{(S^n)} := \{\sum_{i=1}^m g_i x_i \in G^{(S^n)} \mid \sum_{i=1}^m g_i = 0\}$ give an Eilenberg-MacLane spectrum, which will allow us to define configuration cohomology $H_{\text{conf}}^n(X; G)$ as $[X, G_0^{(S^n)}]$. The duality between $H_*^{\text{conf}}(X; G)$ and $H_{\text{conf}}^*(X; G)$ can be concisely stated as “elements of the former are represented by maps from spheres to formal sums over X , while elements of the latter are represented by maps from X to formal sums over spheres”.

Finally, we will explore how configuration cohomology can be used to represent cohomology operations. Notably, we will give a geometric representation of the cup product $\smile: H^n(X; R) \times H^m(X; R) \rightarrow H^{n+m}(X; R)$ and, conditioned on a conjecture, of the Bockstein homomorphism $H^n(X; \mathbb{Z}_2) \rightarrow H^{n+1}(X; \mathbb{Z}_2)$ for the sequence $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2$, which happens to be the Steenrod Square Sq^1 [4, p. 489]. We will also generalize this construction to represent Bockstein homomorphisms for the sequences $\mathbb{Z}_p \rightarrow \mathbb{Z}_{p^2} \rightarrow \mathbb{Z}_p$.

1.1 Conventions and Prerequisites

This thesis is written to be understandable by a graduate student with an interest in topology. We will assume an understanding of the basics of point-set topology (Hausdorff spaces, Compact spaces, the pasting lemma etc.) and algebraic topology (CW Complexes, Singular Homology, the Eilenberg-Steenrod axioms, relative homotopy groups, Singular Cohomology etc.). We will also assume an understanding of basic category theory (Functors, Natural Transformations, (Co)Limits, Adjoints, and the Yoneda Lemma). The thesis will not require an understanding of ∞ -categories or stable homotopy theory, whichever aspects of these theories we use we will explain fully.

Throughout this thesis, due to some technical disadvantages of the category of topological spaces, we will be working in a different category. Specifically, we will define the notion of a k -space:

Definition 1.1 (Compactly Generated Space, k -space). A topological space X is *Compactly Generated* if a subset $U \subset X$ is open if and only if $U \cap K$ is open in K for all compact subsets K of X . If X is also Hausdorff, then it is called a k -space. Given a Hausdorff space X , its k -ification $k(X)$ is the topological space whose underlying set is X , and whose topology consists of all sets U such that $U \cap K$ is open in K for all compact subsets K in X . $\triangle$

For this thesis, we will be working in the full subcategory **kTop** of topological spaces whose objects are k -spaces. The disadvantage of this category is that Hausdorffness is only preserved by subspaces, not quotients, and similarly the property of being compactly generated is only preserved by quotients, not subspaces, so we will have to stay vigilant not to leave this category during our constructions. In practice, this will not cause an issue, and the advantages of **kTop** will far outweigh this inconvenience. Namely, every CW-complex is a k -space, and the product of two CW-complexes in the category of k -spaces itself has a natural CW-structure, unlike the product in the category of topological spaces, which might have too coarse of a topology. [6, p. 149-150] Similarly, if we take the geometric realization of the product of two simplicial sets, it will be different than if we took geometric realizations first, and then took their product – except if we take this product in the category of k -spaces. [4, p. 536-539] The way to pass from the topological product of two k -spaces X and Y to the product in the category of k -spaces is to take the k -ification $k(X \times Y)$. For an arbitrary Hausdorff space Z , the induced continuous map $k(Z) \rightarrow Z$ that sends each point to itself is a weak homotopy equivalence (this can be seen from the fact that maps from spheres $S^n \rightarrow Z$ as well as maps $S^n \times I \rightarrow Z$ that witness homotopies map into a compact subspace of Z , on which k -ification does nothing), so for the purposes of algebraic topology, Z can be replaced with $k(Z)$. To summarize, we will work with **kTop** because the product is topologically nicer behaved for our purposes, without losing any important homotopy-theoretic information in the process. For the rest of this thesis, every topological space is a k -space unless stated otherwise, or unless it needs to be proven.

2 Configuration Homology

In this section, we will define Configuration Homology and prove that it is a Homology Theory. In the process, we will develop the notion of decomposable maps, which we will also use later in this thesis. We start with the following definition:

Definition 2.1 (Formal Sum Space). If X is a topological space, and G is a discrete abelian group, the *pre-sum space* of X over G is

$$G(X) := \prod_{n=1}^{\infty} (G \times X)^n.$$

The *formal sum space* is a topological space with the underlying set $G^{(X)} := \bigoplus_{x \in X} G$. We have a canonical map $q : G(X) \rightarrow G^{(X)}$ sending each $(g_i, x_i)_{i=1}^n$ to $\sum_{i=1}^n g_i x_i$. We define the topology on $G^{(X)}$ by requiring that q is a quotient map. In the special case where X is the empty set, $G^{(X)}$ is the trivial group $\{0\}$. $\triangle$

Note that this is an abelian topological group, as concatenation in $G(X)$ is continuous. Also note that this is functorial in both X and G , and this functor preserves homotopies (as any homotopy $X \times I \rightarrow Y$ ascends to one on $G(X) \times I \rightarrow G(Y)$ and then descends to one on $G^{(X)} \times I \rightarrow G^{(Y)}$).

To be consistent with our conventions established in Section 1.1, we need to show that $G^{(X)}$ is a k -space: Being compactly generated is preserved by quotients, so we only need to show that $G^{(X)}$ is Hausdorff. Since $G^{(X)}$ is a topological group, this amounts to showing that $0 \in G^{(X)}$ is a closed point. This is equivalent to $q^{-1}(0) \subset G(X)$ being closed. Focusing in on a component $C := \prod_{i=1}^n \{g_i\} \times X$ of $G(X)$ (where $g_i \in G$), and treating it as just X^n by ignoring the g_i coordinates, we see that $C \cap q^{-1}(0)$ is a finite union of partial diagonals (sets of the form $\{(x_1, \dots, x_n) \in X^n \mid \forall i, j \in I, x_i = x_j\}$, where $I \subset \{1, \dots, n\}$). Each of these partial diagonals is closed by the Hausdorffness of X , so $G^{(X)}$ is a k -space as desired.

This allows us to define configuration homology.

Definition 2.2 (Configuration Homology). Given a pair (X, A) of topological spaces, where A is a subspace of X , and an abelian group G , we define *configuration homology* as $H_n^{\text{conf}}(X, A; G) := \pi_n(G^{(X)}, G^{(A)}, 0)$. The long exact sequence in homology is given by the long exact sequence of relative homotopy groups. $\triangle$

Given these definitions, we will prove that Configuration Homology is a homology theory in the following way: First, we will construct a morphism from singular homology to configuration homology, and then show that it is an isomorphism. The key idea for that step will be showing that when $G^{(X)}$ has a certain local contractability condition, then any element of $\pi_n(G^{(X)})$ has a representative $f : D^n \rightarrow G^{(X)}$ that lifts to a map $D^n \rightarrow G(X)$ (note that f needs to map to 0 on ∂D^n , but its lift need not be constant on ∂D^n). This

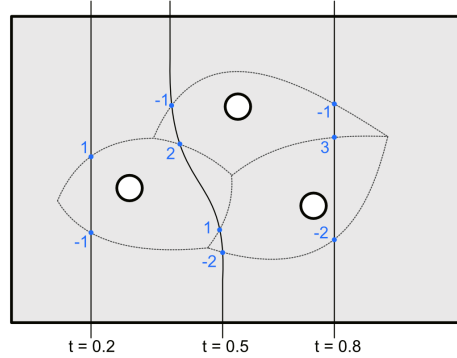


Figure 1: An example of a loop in the formal sum space of a rectangle with three holes in it, with coefficients in $\mathbb{Z}$. The loop starts and ends at 0, but along the way it splits up into different summands, and then merges back together.

property will be called decomposability, and establishing it will be the main focus of the “Decomposable Maps” section. We will then show that any such decomposable map is homotopic to something that comes from our map from singular homology (proving surjectivity), and then that any null-homotopy $D^n \times I \rightarrow G^{(X)}$ (relative to ∂D^n) of such decomposable maps can also be homotoped to something that comes from singular homology (proving injectivity).

Of course, none of this is relevant if that “certain local contractability condition” does not actually hold. Proving it does hold will be broken down into the final two subsections: first, we will show that $G^{(X)}$ has the desired condition if we assume a niceness condition on X . Second, we will show X has this niceness condition whenever it is the geometric realization of a simplicial set. Since all CW-complexes are homotopy equivalent to such a space, this will prove the theorem for all CW complexes.

2.1 Constructing an Isomorphism From Singular Homology to Configuration Homology

In this section, we will construct a morphism from $H_*^{\text{sing}}(X; G)$ to $H_*^{\text{conf}}(X; G)$. We do this by first constructing a chain complex $C_*^{\text{conf}}(X; G)$ whose homology groups are $H_*^{\text{conf}}(X; G)$, and then giving an explicit morphism of chain complexes $C_*^{\text{sing}}(X; G) \rightarrow C_*^{\text{conf}}(X; G)$.

We begin our construction by labelling the vertices of Δ^n from 0 to n for all n . Let $i_n : \Delta^n \rightarrow \Delta^{n+1}$ be the face map sending each vertex k to $k+1$. We define

$$C_n^{\text{conf}}(X; G) := \{f : \Delta^n \rightarrow G^{(X)} \mid f(\Lambda_0^n) = 0\},$$

where Λ_0^n is the union of all faces of Δ^n except $d_0 \Delta^n$. This is an abelian group under pointwise addition. The boundary maps in this chain complex are

$\partial_n : C_n^{\text{conf}}(X; G) \rightarrow C_{n-1}^{\text{conf}}(X; G)$ sending f to $f \circ i_{n-1}$. The composition of two boundary maps is zero, as Λ_0^n and Δ^n share all $(n-2)$ -faces. Furthermore, the homology groups of this chain complex are the homotopy groups of $G^{(X)}$: this is because $\ker(\partial_n)$ consists of maps $f : \Delta^n \rightarrow G^{(X)}$ sending the boundary to 0, which represent elements in $\pi_n(G^{(X)})$. An element is in $\text{im}(\partial_{n+1})$ if and only if it can be homotoped to 0 relative to the boundary, which is exactly the relation defining equivalence classes in $\pi_n(G^{(X)})$. Thus the homology groups of $C_*^{\text{conf}}(X; G)$ are indeed $H_*^{\text{conf}}(X; G)$ as we defined it earlier.

Now, let $B(\Delta^n)$ be the barycentric subdivision of Δ^n . Vertices in this subdivision can be labelled by non-empty subsets of $\{0, \dots, n\}$, and a k face by a sequence of strict inclusions of length $k+1$. Let $w_n : \Delta^n \rightarrow \Delta^n = B(\Delta^n)$ be the map induced by sending each vertex k to $\{k, k+1, \dots, n\}$. Note that $w_n \circ i_{n-1} = i_{n-1} \circ w_{n-1}$, as for any vertex $k \in \Delta^{n-1}$, they both evaluate to the vertex $\{k+1, \dots, n\}$.

Another piece of necessary setup is the following: There is a natural action of the symmetric group Σ_{n+1} on Δ^n by permuting vertices. We also need to use the function $\text{sign} : \Sigma_n \rightarrow \{-1, 1\}$ which sends an element to 1 if and only if it is in the alternating group A_n .

This allows us to finally define the morphism $W : C_*^{\text{sing}}(X; G) \rightarrow C_*^{\text{conf}}(X; G)$: on each generator $gf \in C_*^{\text{sing}}(X; G)$ with $g \in G$ and $f : \Delta^n \rightarrow X$, we define for $x \in \Delta^n$

$$W(gf)(x) = \sum_{\sigma \in \Sigma_{n+1}} \text{sign}(\sigma) gf(\sigma w_n(x)).$$

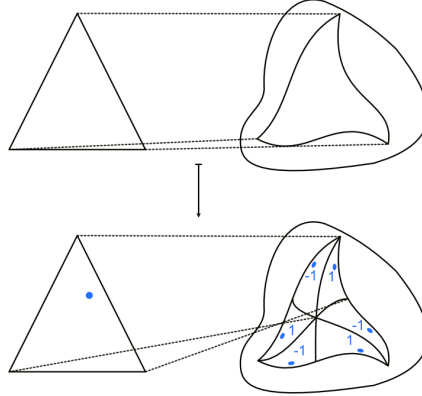


Figure 2: A visual representation of the map W . If the coefficient group is $\mathbb{Z}$, this picture shows how W turns the chain $f : \Delta^2 \rightarrow X$ (top map) into a map $\Delta^2 \rightarrow \mathbb{Z}^{(X)}$ (bottom map).

To confirm that this is in fact a valid definition, we first need to show that $W(gf)(\Lambda_0^n) = 0$. So, suppose $x \in \Lambda_0^n$. Then $w_n(x)$ is in an $(n-1)$ -face of $B(\Delta^n)$ of the form

$$\{0, \dots, n\} \supset \{1, \dots, n\} \supset \dots \supset \{n\},$$

but with one set skipped, which cannot be the set $\{0, \dots, n\}$. So, suppose $\{k, \dots, n\}$ is skipped. Then there are precisely two elements in Σ_{n+1} that act as the identity on this face: the identity, and the element that exchanges k and $k-1$. One of them is in A_{n+1} , and the other is not. Extrapolating, this means that for any $\sigma \in \Sigma_{n+1}$, there exists a unique $\sigma' \in \Sigma_{n+1}$ such that $\sigma(w_n(x)) = \sigma'(w_n(x))$ and $\text{sign}(\sigma) = -\text{sign}(\sigma')$. This gives us

$$W(gf)(x) = \sum_{\sigma \in A_{n+1}} \text{sign}(\sigma)gf(\sigma w_n(x)) + \text{sign}(\sigma')gf(\sigma' w_n(x)) = 0$$

as desired. We also need to confirm is that W commutes with boundary maps. First, let $\Sigma_{n+1}(k) \subset \Sigma_{n+1}$ be the subgroup of Σ_{n+1} consisting of all elements that fix k . Then we have

$$W(\partial_n(gf))(x) = \sum_{i=0}^n (-1)^i \sum_{\sigma \in \Sigma_n} \text{sign}(\sigma)gf|_{d_i \Delta^n}(\sigma w_{n-1}(x))$$

Where $f|_{d_i \Delta^n}$ is to be interpreted as f precomposed with the face map $[0, \dots, n-1] \rightarrow [0, \dots, n]$ that skips i . For $i=0$, this is just the map i_{n-1} , and for all other i , we can write the face map as $s_i i_{n-1}$ with $s_i = (012 \dots i)^{-1} \in \Sigma_{n+1}$. We can therefore rewrite the above formula as

$$\begin{aligned} W(\partial_n(gf))(x) &= \sum_{i=0}^n (-1)^i \sum_{\sigma \in \Sigma_n} \text{sign}(\sigma)gf(s_i i_{n-1}(\sigma w_{n-1}(x))) \\ &= \sum_{i=0}^n (-1)^i \sum_{\sigma \in \Sigma_{n+1}(0)} \text{sign}(\sigma)gf(s_i \sigma i_{n-1}(w_{n-1}(x))) \\ &= \sum_{i=0}^n \sum_{\sigma \in \Sigma_{n+1}(0)} \text{sign}(s_i \sigma)gf(s_i \sigma w_n(i_{n-1}(x))) \\ &= \sum_{\sigma \in \Sigma_{n+1}} \text{sign}(\sigma)gf(\sigma w_n(i_{n-1}(x))) = \partial_n(W(gf))(x) \end{aligned}$$

as desired. In the following sections, we will show that W induces an isomorphism on homology.

2.2 Decomposable Maps

This section is the technical heart of this proof. We will develop a general notion of decomposability, and then use the results to show that, under certain conditions, the morphism $W_* : H_*^{\text{sing}}(X) \rightarrow H_*^{\text{conf}}(X)$ we just constructed is an isomorphism.

We start with a pair of definitions

Definition 2.3 (Quasi-Involutive Topological Magma). A *topological magma* is a topological space M equipped with an operation $\star : M \times M \rightarrow M$ called composition. If M is also equipped with a map $^- : M \rightarrow M$ called a quasi-involution, then it is called *quasi-involutive*. There are no algebraic properties these operations need to satisfy. A map between quasi-involutive topological magmas is called a homomorphism if it preserves both operations. $\triangle$

Note that every topological group is a quasi-involutive topological magma, with composition being group composition, and the quasi-involution being the inverse-map.

Definition 2.4 (Decomposition Structure). Let G be a topological group. A *decomposition structure* on G is a quotient map $M \rightarrow G$ from a quasi-involutive topological magma M . Furthermore, we also require this quotient map to be a homomorphism. $\triangle$

The most important example of a decomposition structure for the purposes of this thesis is the map $q : G(X) \rightarrow G^{(X)}$ giving a decomposition structure on $G^{(X)}$. The magma structure on $G(X)$ is given by concatenation, and the quasi-involutive structure is given by applying the inverse map on G to all of the G coordinates.

This example is the motivation for the term “decomposition”: suppose Y is connected, and there is a map $f : Y \rightarrow G^{(X)}$ that factors through q as, say, $f = qf'$. Then f' maps into one of the components $\prod_{i=1}^n \{g_i\} \times X$ of $G(X)$, which means f can be decomposed as a sum of maps $\sum_{i=1}^n g_i f_i$, where each f_i is a map from Y to X . In other words, maps into $G^{(X)}$ that factor through the decomposition structure can be decomposed into a sum of maps that are easier to analyse.

This motivates the following definition:

Definition 2.5 (Decomposable Map, Decomposable Homotopy). Let $q : M \rightarrow G$ be a decomposition structure on a topological group G . A map $f : X \rightarrow G$ is called *decomposable* if there exists a continuous function $l_f : X \rightarrow M$ such that $f = q \circ l_f$. The function l_f is then called a *decomposition* of f . If $f_1, f_2 : X \rightarrow G$ are decomposable maps, a *decomposable homotopy* between them is a homotopy $h : X \times I \rightarrow G$ from f_1 to f_2 that is also a decomposable map.

More generally, if $A \subset X$ is some subspace, a map $f : X \rightarrow G$ that is constant on A is called *relatively decomposable* with respect to A if the induced map $X/A \rightarrow G$ is decomposable. Similarly, a homotopy relative to A is relatively decomposable if the induced map $X/A \times I \rightarrow G$ is decomposable.

The class of all decomposable maps from X to G is denoted as $D(X, G)$. The class of all decomposable maps from X to G relative to A is denoted $D(X, A, G)$. The analogous classes for decomposable homotopies are $DH(X, G)$ and $DH(X, A, G)$. $\triangle$

Decomposable maps are a particularly nice class of maps, and the remainder of this section aims to establish some important formal properties of them, which we will be able to exploit greatly. First, some immediate properties:

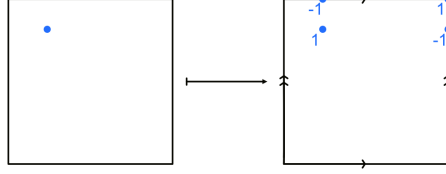


Figure 3: An example of a decomposable map from I^2 to $\mathbb{Z}(T^2)$ with respect to the decomposition structure $\mathbb{Z}(T^2) \rightarrow \mathbb{Z}(T^2)$, where T^2 is the torus. This map is decomposable, as we can write it up as the sum of the identity and the constant map to the corner, minus the two projection maps to the sides. However, note that this map factors through the quotient $I^2 \rightarrow S^2$ collapsing the boundary. The resulting map $S^2 \rightarrow \mathbb{Z}(T^2)$ is not decomposable, as the decomposition would have to map to infinitely many points at the basepoint of S^2 .

Lemma 2.6. *The classes $D(X, G)$, $D(X, A, G)$, $DH(X, G)$, and $DH(X, A, G)$ all contain all constant maps, and are closed under group composition and inverses. The latter two contain all constant homotopies of decomposable maps, and are closed under composition of homotopies and taking reverse homotopies.*

Proof. The following proofs work for all 4 classes: All constant maps are decomposable by the surjectivity of $q : M \rightarrow G$. Furthermore, for any pair of decompositions l_{f_1}, l_{f_2} of f_1 and f_2 respectively, the composition $l_{f_1} \star l_{f_2}$ defines a decomposition of $f_1 \cdot f_2$. For any decomposition l_f of some f , taking the quasi-involution $\bar{l}_f$ gives a decomposition of f^{-1} .

Now, for the proofs that are specific to homotopies: If l_f is a decomposition of $f : X \rightarrow G$, then $l_h : (x, t) \mapsto l_f(x)$ defines a decomposition of the constant homotopy of f . If $h_1, h_2 : X \times I \rightarrow G$ are decomposable homotopies which can be composed as homotopies, then $h_1 * h_2$ is decomposable for the following reason: First, we can pointwise left-compose by the inverse of the constant homotopy of $h_1(x, 1) = h_2(x, 0)$ to reduce to the case where $h_1 * h_2(x, \frac{1}{2}) = e$. A decomposition of this is given by

$$l_{h_1 * h_2}(x, t) := \begin{cases} l_{h_1}(x, 2t) \star l_{h_2}(x, 0) & \text{if } t \leq \frac{1}{2} \\ l_{h_1}(x, 1) \star l_{h_2}(x, 2t - 1) & \text{otherwise.} \end{cases}$$

Finally, for reverse homotopies, just precompose our decomposition with the map $(x, t) \rightarrow (x, 1 - t)$. All these constructions also work for the relative cases. $\square$

This already has some nice consequences:

Corollary 2.7. *Let $f : I^n \rightarrow G$ be a map such that for some N , f is decomposable when restricted to the set $[\frac{k_1}{N}, \frac{k_1+1}{N}] \times \dots \times [\frac{k_n}{N}, \frac{k_n+1}{N}]$ for all $(k_1, \dots, k_n) \in \{0, \dots, N-1\}^n$. Then f is decomposable.*

To prove this, we just apply the fact that the composition of decomposable homotopies is decomposable over and over again.

There is another more complicated corollary, for which we need the following definition:

Definition 2.8 (Pyramids). Let F be one of the faces of the hypercube I^n . The *pyramid* P_F is defined to be the set $\{(1-t)(x_1, \dots, x_n) + t(\frac{1}{2}, \dots, \frac{1}{2}) \in I^n \mid t \in I, (x_1, \dots, x_n) \in F\}$. $\triangle$

Note that I^n is the union of all its pyramids.

Corollary 2.9. *Let $f : I^n \rightarrow G$ be a map that is decomposable when restricted to each pyramid of I^n . Then f is decomposable.*

Proof. First, we will subdivide the cube I^n into cubes of the form $\prod_{i=1}^n [\frac{k_i}{2}, \frac{k_i+1}{2}]$ where each $k_i \in \{0, 1\}$. By the previous corollary it is enough to prove the statement on these cubes, and by symmetry we can focus on $[0, \frac{1}{2}]^n$. We normalise this to $[0, 1]^n$. Now, the pyramids in the original cube restrict here to sets of the form $J_F := \{t(x_1, \dots, x_n) + (1-t)(1, \dots, 1) \mid t \in I, (x_1, \dots, x_n) \in F\}$, where F is a face of I^n that does not contain the vertex $(1, \dots, 1)$. The union of all such faces is called the corner of I^n , and is labelled C^{n-1} . We can construct a quotient map $r_n : C^{n-1} \times I \rightarrow I^n$ by sending each (x, t) to $((1-t)x + t(1, \dots, 1))$. This quotient map effectively just collapses the set $C^{n-1} \times \{1\}$. Note that the sets $F \times I$ (with F one of the faces in C^{n-1}) map to the pyramids. This means we have reduced our question to the following: given a map $f' : C^n \times I \rightarrow G$, if $f'|_{F \times I} \in D(F \times I, F \times \{1\}, G)$ for all faces $F \subset C^n$, then is $f' \in D(C^n \times I, C^n \times \{1\}, G)$?

We will actually prove a stronger statement instead: We will show that for any $k \geq 1$ and map $f' : C^n \times I^k \rightarrow G$ such that $f'|_{F \times I^k} \in D(F \times I^k, F \times I^{k-1} \times \{1\}, G)$ for all faces F in C^n , then $f' \in D(C^n \times I^k, C^n \times I^{k-1} \times \{1\}, G)$.

On this, we proceed inductively on n : for $n = 0$, C^n is just a point, so our statement holds. For the inductive step, note that we have a homeomorphism $A : C^{n-1} \times I \sqcup_{C^{n-1} \times \{0\}} I^n \rightarrow C^n$ sending each $((x_1, \dots, x_n), x_{n+1}) \in C^{n-1} \times I$ to $(x_1, \dots, x_{n+1})$ and each $(x_1, \dots, x_n) \in I^n$ to $(x_1, \dots, x_n, 0)$ (note that this is effectively just a pushout diagram, as illustrated in Figure 4). The faces $F \subset C^n$ under A correspond to either $I^n \subset C^{n-1} \times I \sqcup_{C^{n-1} \times \{0\}} I^n$ or an $F' \times I \subset C^{n-1} \times I \sqcup_{C^{n-1} \times \{0\}} I^n$ for some face $F' \subset C^{n-1}$. This means we can write the map f' as a map

$$f' : C^{n-1} \times I^{k+1} \sqcup_{C^{n-1} \times I^k \times \{0\}} I^{n+k} \rightarrow G$$

and by induction we can assume it is decomposable (in a relative way) on both $C^{n-1} \times I^{k+1}$ and I^{n+k} .

The next step is to construct a homeomorphism $C^{n-1} \times I^{k+1} \sqcup_{C^{n-1} \times I^k \times \{0\}} I^{n+k} \rightarrow I^{n+k}$ that sends $C^{n-1} \times I^{k+1}$ to one half of I^{n+k} and sends I^{n+k} to the other half, thus translating the question into “is the composition of two (relatively) decomposable homotopies (relatively) decomposable”, which we know is

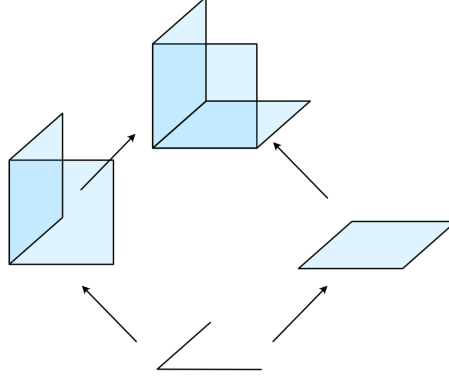


Figure 4: An illustration of the pushout diagram that induces the map A in the proof of Corollary 2.9 for $n = 2$.

true. We can reduce to the case of $k = 0$ (by just multiplying our homeomorphism by the identity on I^k), in which case we need a homeomorphism

$$h : C^n \rightarrow I^n$$

that sends one face to one half of I^n , and all other faces to the other half.

We will now construct h using a series of observations (the following construction is visualized in Figure 5). First, for $n = 1$, the construction is trivial: h just sends one face (of C^1) to one half (of I) and the other face to the other half. For higher n , we start our construction by treating C^n as a subspace of $\mathbb{R}^{n+1}$, and then radially projecting C^n flat onto the plane $\mathbb{R}^n \times \{0\}$ from the point $(1, 1, \dots, 1, 2)$. This gives a homeomorphism $C^n \rightarrow [-1, 1]^n$ that sends one of the faces of C^n to $[0, 1]^n$. Our goal is then to find a self-homeomorphism φ_n of $[-1, 1]^n$ that sends $[0, 1]^n$ to $[0, 1] \times [-1, 1]^{n-1}$ (as, up to normalizing, this would give us the desired map h). We will construct this inductively: first, we will construct a self-homeomorphism φ_2 on $[-1, 1]^2$ that sends $[0, 1]^2$ to $[0, 1] \times [-1, 1]$. Then, for any $n > 2$, we will construct $\varphi_n : [-1, 1]^n \rightarrow [-1, 1]^n$ the following way: first, we will apply φ_{n-1} to the last $n - 1$ coordinates (and the identity to the first one). Then we apply φ_2 to the first two coordinates (and again the identity to the remaining ones). This will first send $[0, 1]^n$ to $[0, 1]^2 \times [-1, 1]^{n-2}$, then to $[0, 1] \times [-1, 1]^{n-1}$ as desired. So, all we have to do is construct φ_2 . We do this by first applying the homeomorphism $[-1, 1]^2 \rightarrow D^2$ which scales down appropriately along radial lines from the origin. If we write elements of D^2 in polar coordinates, we can describe a self-homeomorphism on D^2 by sending (r, θ) to $(r, 2\theta)$ if $0 \leq \theta < \frac{\pi}{2}$, and sending it to $(r, \frac{2}{3}(\theta - \frac{\pi}{2}) + \pi)$ if $\frac{\pi}{2} \leq \theta < 2\pi$. Applying this self-homeomorphism and then transforming back to $[-1, 1]^2$ gives a self-homeomorphism on $[-1, 1]^2$. We can manually check that this is a valid candidate for φ_2 , which completes the proof. $\square$

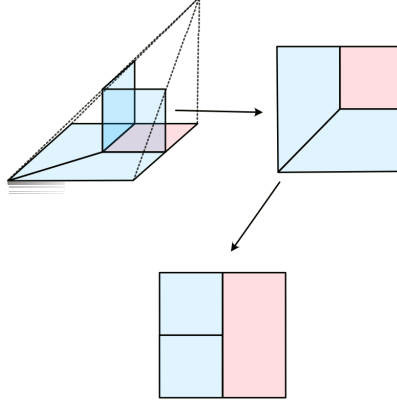


Figure 5: An illustration of the homeomorphism $h : C^n \rightarrow I^n$ we construct in the proof of Corollary 2.9 for $n = 2$.

There is one more property of decomposable maps we need to show, and then we have enough formal properties to proceed further.

Lemma 2.10. *If X is contractible, then a null homotopy $h : X \times I \rightarrow G$ is decomposable if and only if $h' : X \times I/X \times \{1\} \rightarrow G$ defined as*

$$h'(x, t) := \begin{cases} h(x, 2t) & \text{if } t \leq 0.5, \\ h(x, 1) & \text{otherwise} \end{cases}$$

is decomposable.

Proof. If h' is decomposable, we can just precompose that decomposition with the map $X \times I \rightarrow X \times I, (x, t) \mapsto (x, \frac{1}{2}t)$ to get a decomposition of h . In the other direction, if l_h is a decomposition of h and $r : X \times I \rightarrow X$ is a contraction, then we can define

$$l_{h'}(x, t) := \begin{cases} l_h(x, 2t) & \text{if } t \leq 0.5 \\ l_h(r(x, 2t - 1), 1) & \text{otherwise.} \end{cases}$$

which is a decomposition of h' . □

After the following definition, we will be able to use these properties to prove two big results in a row:

Definition 2.11 (Locally Decomposably Contractible). A topological group with a decomposition structure $q : M \rightarrow G$ is called *locally decomposably contractible* with respect to q , if for any $g \in G$ and open $U_g \ni g$, there exists a smaller open $g \in V_g \subset U_g$ such that there exists a contraction $r : V_g \times I \rightarrow V_g$ which has the property that for all decomposable maps $f : X \rightarrow V_g$, the induced homotopy $r_f : X \times I \rightarrow V_g$ is also decomposable. △

The first big result is the following:

Theorem 2.12. *Let $f : I^n \rightarrow U \subset G$ be a map (where U is open in G) and let $h : \partial I^n \times I \rightarrow U$ be a homotopy from $f|_{\partial I^n}$ to a map that is decomposable on each face individually. If G is locally decomposably contractible, then h can be extended to a homotopy on I^n from f to a decomposable map.*

Proof. We proceed by induction on n . On $n = 0$, the statement is trivial (as all constant maps are decomposable), so assume $n > 0$. What we have then essentially is the following: Let $J^n \subset \partial I^{n+1}$ be the union of all faces except $F^n = I^n \times \{1\}$. We are given a map $h : J^n \rightarrow U \subset G$ that is decomposable on all faces of $J^n \cap F^n$. We want to extend this to a map $h' : I^{n+1} \rightarrow U$ that is decomposable on F^n .

Let $r_n : I^{n+1} \rightarrow J^n$ be a retract (it exists by the homotopy extension property). Subdivide I^n into cubes $K_{k_1, \dots, k_n} := \prod_{i=1}^n [\frac{k_i}{N}, \frac{k_i+1}{N}]$ with $k_i \in \{0, \dots, N-1\}$, and assume N is large enough such that $r_n(K_{k_1, \dots, k_n} \times [\frac{N-1}{N}, 1])$ maps into (via h) some decomposably contractible open $U_{k_1, \dots, k_n} \subset U$. For a general m -face E for $m < n+1$ of some cube $K_{k_1, \dots, k_n} \times [\frac{N-1}{N}, 1]$, we define U_E as the intersection of all the $U_{k_1, \dots, k_n}$ s that contain $h(r_n(E))$.

The general construction of h' is then the following (for a visual aide to this construction, see Figure 6): $h'|_{I^n \times [0, \frac{N-1}{N}]} := h \circ r_n|_{I^n \times [0, \frac{N-1}{N}]}$ (and of course, $h'|_{J^n} := h$). On the strip $I^n \times [\frac{N-1}{N}, 1]$, we do the construction by induction on skeletons: first, on any 1-face that is not in F^n , and on which h' is not already defined, we define it to just be $h \circ r_n$. For any $1 < m \leq n$, on an m -face E that is not in F^n and on which h' is not already defined, we know by induction the following things: first, h' is defined on all $m-1$ -faces of E except for the one on F^n , second, h' maps inside U_E on those faces, and third, restricted to $F^n \cap E$, h' is decomposable on all the $(m-2)$ -faces. Thus, we can use our inductive hypothesis to extend h' to E such that $h'(E) \subset U_E$ and $h'|_{E \cap F^n}$ is decomposable.

By the end of this construction, the only faces on which h' is not defined are the $K_{k_1, \dots, k_n} \times [\frac{N-1}{N}, 1]$ s and their corresponding faces on F^n . However, note that the question of whether we can extend h' to these cubes is just a zoomed in version of whether we can extend h to I^{n+1} , the only difference being that the former we know maps into a decomposably contractible open set, while the latter maps into just some open set.

With this reduction in mind, we forget about our subdivision and just assume that the open U that h maps into is decomposably contractible from the beginning. Then all we need to do is show that $h|_{J^n \cap F^n}$ can be extended to a decomposable map on F^n , as we can further extend h to all of I^{n+1} by the contractibility of U .

This means we have now reduced to the following problem: given a map $h : \partial I^n \rightarrow U \subset G$ that is decomposable on all the faces, if U is decomposably contractible, then h can be extended to a decomposable map $h' : I^n \rightarrow U$. Let $r : U \times I \rightarrow U$ be a decomposable contraction. Let E be a face of I^n . Then the map $h_E : E \times I/E \times \{1\} \rightarrow U$ defined as

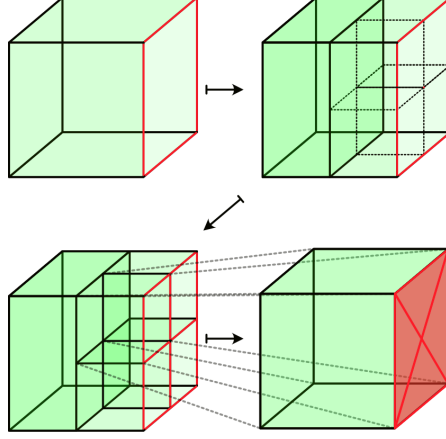


Figure 6: An illustration of the construction of h' in the proof of Theorem 2.12. The green regions are places where h' has already been defined, the red regions represent areas on which we know h' is decomposable.

$$h_E(x, t) := \begin{cases} r(h(x), 2t) & \text{if } t \leq 0.5 \\ r(h(x), 1) & \text{otherwise.} \end{cases}$$

is decomposable by Lemma 2.10. Letting $p_E : E \times I/E \times \{1\} \rightarrow P_E, (x, t) \mapsto ((1-t)x + t(\frac{1}{2}, \dots, \frac{1}{2}))$ be the homeomorphism that identifies $E \times I/E \times \{1\}$ with the pyramid of the E face, we can define h' via the rule $h'|_{P_E} = h_E \circ p_E^{-1}$. This is well defined, continuous, and decomposable on all of the pyramids, so, by Corollary 2.9, h' is decomposable as a whole. This completes the proof. $\square$

The second big result is a direct application of this to formal sum spaces:

Theorem 2.13. *Given a CW-complex X and a discrete abelian group G , if $G^{(X)}$ is locally decomposably contractible (with respect to its defining decomposition structure), then $W_* : H_n^{\text{sing}}(X; G) \rightarrow H_n^{\text{conf}}(X; G)$ is an isomorphism.*

Proof. The chain-map $W : C_*^{\text{sing}}(X; G) \rightarrow C_*^{\text{conf}}(X; G)$ can be written as a composition of maps $C_*^{\text{sing}}(X; G) \twoheadrightarrow \text{im}(W) \hookrightarrow C_*^{\text{conf}}(X; G)$, and we will show both of these induce isomorphisms on homology. We start with the second one, which we will call $f : \text{im}(W) \hookrightarrow C_*^{\text{conf}}(X; G)$. First, we will show $f_* : H_n(\text{im}(W)) \rightarrow H_n^{\text{conf}}(X; G)$ is surjective: let $\alpha : \Delta^n \rightarrow G^{(X)}$ represent an n -cycle in $C_*^{\text{conf}}(X; G)$ (i.e. assume α is 0 on the boundary). Then by Theorem 2.12, we can homotope α relative to the boundary to a decomposable map (note that Theorem 2.12 applies to cubes, not simplices. This is not relevant for our purposes, as they are homeomorphic). Without loss of generality, assume α is already decomposable. This means $\alpha = \sum_{i=1}^m g_i \alpha_i$ with $g_i \in G$ and $\alpha_i : \Delta^n \rightarrow X$. Let $\alpha' := \sum_{i=1}^m g_i \alpha_i \in C_*^{\text{sing}}(X; G)$. Now, we claim $W(\alpha')$ is homotopic to

α relative to the boundary, which is all we need to show that f_* is surjective, as the injectivity of f already implies that $W(\alpha')$ is an n -cycle in $\text{im}(W)$. To construct this homotopy, we first explicitly write

$$W(\alpha')(x) = \sum_{\sigma \in \Sigma_{n+1}} \text{sign}(\sigma) \alpha(\sigma w_n(x)).$$

In order to homotope this to α , we homotope the σw_n s themselves. Since these homotopies will not respect the action of σ , we will define $w_{n,\sigma} := \sigma w_n$ to avoid confusion. Let non-empty subsets of $\{0, \dots, n\}$ represent vertices in the barycentric subdivision of Δ^n . If $v \subset \{0, \dots, n\}$ is of the form $\{k, k+1, \dots, n\}$ for $0 \leq k \leq n$, then we define $L_v : I \rightarrow \Delta^n, t \mapsto (1-t)\{k, \dots, n\} + t\{k\}$. If v is not of this form, we define $L_v : I \rightarrow \Delta^n$ to be the constant v . With this in mind, we can define $h_{w_n,\sigma} : \Delta^n \times I \rightarrow \Delta^n$ as

$$h_{w_n,\sigma}((x_0, \dots, x_n), t) := \sum_{i=0}^n x_i L_{w_{n,\sigma}(i)}(t).$$

These homotopies agree with each other on the boundaries in the sense that if $w_{n,\sigma_1}(x) = w_{n,\sigma_2}(x)$, then $h_{w_{n,\sigma_1}}(x, t) = h_{w_{n,\sigma_2}}(x, t)$ for each t . This means that $h : \Delta^n \times I \rightarrow G^{(X)}$ defined as

$$h(x, t) = \sum_{\sigma \in \Sigma_{n+1}} \text{sign}(\sigma) \alpha(h_{w_n,\sigma}(x, t))$$

is a valid homotopy from $W(\alpha')$. Since all points $h_{w_n,\sigma}(x, 1)$ with $\sigma \neq e$ are on $\partial\Delta^n$, on which α evaluates to 0, we have that

$$h(x, 1) = \alpha(h_{w_n,e}(x, 1)) = \alpha(x)$$

as desired.

Before showing that f_* is injective, we will note the following: the chain map $C_*^{\text{sing}}(X; G) \rightarrow C_*^{\text{sing}}(X; G)$ sending a generator $g\alpha$ to

$$\sum_{\sigma \in \Sigma_{n+1}} \text{sign}(\sigma) g\alpha \circ (\sigma w_n)$$

acts as the identity on homology [4, p. 119-124]. We will refer to this as barycentric subdivision. We will use this to, first of all, show that f_* is injective: let α (reusing variables) be a singular n -cycle, and let $\beta : \Delta^{n+1} \rightarrow X$ be 0 on every face except $d_0\Delta^{n+1}$, on which it is $W(\alpha)$. As $W(\alpha)$ is decomposable, by Theorem 2.12, we can homotope β relative to the the boundary to a decomposable map. Assume that β is already decomposable, and let $\beta = \sum_{i=1}^m g_i \beta_i$. Let β' be the singular $n+1$ -chain this sum represents. Now, let α' be the barycentric subdivision of α . Then we have by unwinding definitions that $\partial_{n+1}(W(\beta)) = W(\alpha')$, which proves the injectivity of f_* .

The second half of the proof is showing that $W : C_*^{\text{sing}}(X; G) \rightarrow \text{im}(W)$ is an isomorphism on homology. By the existence of the long exact sequence in

homology, this is equivalent to $\ker(W)$ having only trivial homology groups. To show this, we will introduce a new chain complex we will call $N(X; G)$, which is defined as

$$N_n(X; G) := \left\{ \sum_{i=1}^m g_i \alpha_i \in C_n^{\text{sing}}(X; G) \mid \forall x \in \Delta^n, \sum_{i=1}^m g_i \alpha_i(x) = 0 \in G^{(X)} \right\},$$

with the chain complex structure inherited from $C_*^{\text{sing}}(X; G)$. Now, the standard proof method that shows singular homology is a homology theory can be repurposed to show the functors $X \mapsto H_n(N_*(X; G))$ also give a homology theory (the only interesting Eilenberg-Steenrod axiom here is excision, a proof of which that can be adapted to $H_n(N_*(X; G))$ can be found in [4, p. 119-125]). However, $N_*(\{*\}; G) = 0$, which means the homology of the one-point space in this homology theory is trivial. This implies that $H_n(D_*(X, G)) = 0$ for all n as long as X is a CW-complex.

Performing barycentric subdivision gives a chain map $\ker(W) \rightarrow N_*(X; G)$ (we can see this by unwinding definitions), and there is a natural inclusion $N_*(X; G) \rightarrow \ker(W)$. Composing these two gets us the barycentric subdivision map $\ker(W) \rightarrow \ker(W)$. If this induces an isomorphism on homology, then we are done, as that means the inclusion $N_*(X; G) \rightarrow \ker(W)$ induces a surjection on homology, which means $\ker(W)$ is exact as desired. By the five lemma, it is enough to check that performing barycentric subdivision induces the identity on homology on both $C_*^{\text{sing}}(X; G)$ and $\text{im}(W)$. We know this is true on the former. On the latter, since we already showed the inclusion $\text{im}(W) \hookrightarrow C_*^{\text{conf}}(X; G)$ is an isomorphism on homology, we can check this statement inside $C_*^{\text{conf}}(X; G)$, in which the homotopy h we constructed in the first half of this proof witnesses that barycentric subdivision acts as the identity on $H_*^{\text{conf}}(X; G)$. This completes the proof. $\square$

Now the only thing we need to do is establish that $G^{(X)}$ is locally decomposably contractible for all sufficiently nice spaces X , and that all CW complexes are homotopy equivalent to such a space. The next two sections cover these two steps.

2.3 Ensembles

In this section we will construct a local decomposable contraction on $G^{(X)}$ given a niceness assumption on X . To do this, we will use the following property:

Lemma 2.14. *Let $q : G(X) \rightarrow G^{(X)}$ be the defining decomposition structure of the formal sum space. Then if for some $U \subset G^{(X)}$ a homotopy $h : q^{-1}(U) \times I \rightarrow q^{-1}(U)$ descends to a contraction $h' : U \times I \rightarrow U$, then h' is a decomposable contraction.*

We can prove this by simply applying h to a decomposition of any (decomposable) map $f : Y \rightarrow U$ to get a decomposition of the homotopy $h'(f(y), t)$.

The main issue is that h descending to a homotopy (let alone a contraction) h' is a very strong condition, and without further groundwork it is difficult to make any inductive constructions that do not break this condition.

To overcome this, we will first introduce some notation: For any finite tuple (including the empty tuple) $(g_1, \dots, g_n)$ of elements in G , we will denote the set $\prod_{i=1}^n (\{g_i\} \times X)$ as $E(g_1, \dots, g_n)$, and we will treat it as just an indexed copy of X^n . If $n > 0$, then $E(g_1, \dots, g_n) \subset G(X)$, which means elements of $E(g_1, \dots, g_n)$ represent elements in $G^{(X)}$. In the special case where $n = 0$, we say that the one element of $E(\emptyset) = \{*\}$ represents $0 \in G^{(X)}$. Given a pair of elements $(x_1, \dots, x_n) \in E(g_1, \dots, g_n)$ and $(x'_1, \dots, x'_m) \in E(g'_1, \dots, g'_m)$, we say they are related if they represent the same formal sum in $G^{(X)}$, and we denote this equivalence relation by $\sim$.

We will now break the relation $\sim$ down into a series of relations that are easier to analyse. First, if $\sigma \in \Sigma_n$ is an element of the symmetric group, then $(x_1, \dots, x_n) \in E(g_1, \dots, g_n)$ represents the same element as $(x_{\sigma(1)}, \dots, x_{\sigma(n)}) \in E(g_{\sigma(1)}, \dots, g_{\sigma(n)})$. We denote the relation given by these permutations as $\sim_P$. Second, any element of the form $(x_1, \dots, x_n, x_n) \in E(g_1, \dots, g_{n+1})$ is related to $(x_1, \dots, x_n) \in E(g_1, \dots, g_n, 0)$ (by the definition of $\sim$). We will denote this sub-relation of $\sim$ as $\sim_+$ (note that $\sim_+$ is not reflexive, symmetric, or transitive). Finally, any element of the form $(x_1, \dots, x_{n+1}) \in E(g_1, \dots, g_n, 0)$ is related to $(x_1, \dots, x_n) \in E(g_1, \dots, g_n)$, and we will refer to this sub-relation as $\sim_0$. We can now state the following lemma:

Lemma 2.15. *The relation $\sim$ as described above is generated (as an equivalence relation) by $\sim_P, \sim_+$ and $\sim_0$. Furthermore, if we define $\sim_{P+}$ and $\sim_{P0}$ to be the composition of relations $\sim_P \sim_+$ and $\sim_P \sim_0$ respectively, then for any $l := (x_1, \dots, x_m) \in E(g_1, \dots, g_m)$, there exists a unique-up-to-permutation element $l' := (x'_1, \dots, x'_n) \in E(g'_1, \dots, g'_n)$ with $l \sim l'$ for which the following holds:*

1. *The number n is minimal (i.e. n is the smallest it can be while remaining in the equivalence class of l).*
2. *There exists a pair of (potentially length zero) chains*

$$l =: l_0 \sim_{P+} l_1 \sim_{P+} \dots \sim_{P+} l_k$$

and

$$l_k \sim_{P0} l_{k+1} \sim_{P0} \dots \sim_{P0} l_{m-n} := l'$$

Proof. The second half of the lemma immediately proves the first half, so we will focus on that. We will first describe how to get from l to l' , and then we will show that l' is minimal and unique up to permutation. To get from l to l' , we successively apply $\sim_{P+}$ until we have removed all duplicates in the

X coordinates (each application removes one), and then we apply $\sim_{P_0}$ until there are no more 0s in the G coordinates (each application also removes one). After $m - n$ steps, we arrive at an element $l' \in E(g'_1, \dots, g'_n)$, which has neither duplicates in the X coordinates, nor 0s in the G coordinates.

To show that l' is minimal, we note that the tuple $(x'_1, \dots, x'_n)$ by construction contains all points of X where $\sum_{i=1}^n g'_i x'_i$ is non-zero exactly once, so n cannot be smaller. The uniqueness of l' up to permutation is true by the definition of formal sums. This completes the proof. $\square$

We will be able to use this lemma to make inductive constructions on $G(X)$. We first need to establish some more terminology: Given a subset $U \subset G^{(X)}$, we denote by $E_U(g_1, \dots, g_n)$ the subset of $E(g_1, \dots, g_n)$ containing all elements that represent a formal sum in U . Given a fixed U , we will refer to the disjoint union of $E_U(g_1, \dots, g_n)$ s ranging over all tuples in G as an *ensemble*, which we will denote as E_U . Alternatively, an ensemble is a subspace of $\coprod_{\text{tuples in } G} E(g_1, \dots, g_n)$ that is closed under $\sim$ (by the definition of $\sim$, these are equivalent). Given an ensemble E_U , let $E_{U,n} := E_U \cap \coprod_{\text{tuples in } G \text{ of size at most } n} E(g_1, \dots, g_k)$, which we will refer to as a truncated ensemble. We also define $G_n^{(X)} := \{\sum_{i=1}^k g_i x_i \in G^{(X)} \mid k \leq n\}$. We are now ready to prove the following lemma:

Lemma 2.16. *Let $E_{U,n}$ be a truncated ensemble, and let $h : E_{U,n} \times I \rightarrow E_{U,n}$ be a homotopy that preserves $\sim$, and for which $h(E_{U,n}(g_1, \dots, g_k) \times I) \subset E_{U,n}(g_1, \dots, g_k)$ holds for each tuple. There is a canonical way to extend h to a homotopy $h' : E_{U \cap G_n^{(X)}} \times I \rightarrow E_{U \cap G_n^{(X)}}$ that also preserves $\sim$, and for which $h'(E_{U \cap G^{(X)}}(g_1, \dots, g_k) \times I) \subset E_{U \cap G^{(X)}}(g_1, \dots, g_k)$ also holds.*

Proof. Without loss of generality, assume $U \subset G_n^{(X)}$. We can prove this lemma by induction on the size of the tuple $(g_1, \dots, g_k)$: the homotopy h' is already defined on each $E_U(g_1, \dots, g_k)$ for $k \leq n$, which serves as the base-case for the induction.

For the induction step, assume h' is already defined for each $k \leq n'$ for some n' , and let $(g_1, \dots, g_{n'+1})$ be a tuple in G . Let $(x_1, \dots, x_{n'+1}) \in E_U(g_1, \dots, g_{n'+1})$. By the pigeonhole principle, it is impossible for both each x_i to be distinct, and for each g_i to be non-zero. So, suppose first that, without loss of generality, $x_{n'} = x_{n'+1}$, in which case we can define $h'(x_1, \dots, x_{n'+1}, t)$ to be $h'(x_1, \dots, x_{n'}, t)$ but with the last coordinate duplicated (where $(x_1, \dots, x_{n'}) \in E_U(g_1, \dots, g_{n'} + g_{n'+1})$). If each x_i is distinct, then suppose instead without loss of generality that $g_{n'+1} = 0$. Then we can define $h'(x_1, \dots, x_{n'+1}, t)$ to be $h'(x_1, \dots, x_{n'}, t)$, but with an $x_{n'+1}$ appended to the end (where $(x_1, \dots, x_{n'}) \in E_U(g_1, \dots, g_{n'})$). By Lemma 2.15, this is sufficient to ensure h' preserves $\sim$, which completes the proof. $\square$

Before we can prove the main theorem of this section, we need one final definition:

Definition 2.17 (Symmetrically Hyperlocally Contractible space). A space X is called *symmetrically hyperlocally contractible* if for all n , given the data of

1. A subgroup $S \subset \Sigma_n$
2. An open subset $U \subset X^n$.
3. An open subset $U' \subset \Delta_X^n$ of the fat diagonal (where the fat diagonal Δ_X^n contains all points $(x_1, \dots, x_n)$ in X^n for which $x_i = x_j$ for some $i \neq j$) that is closed under the action of S , and for which we have $U' \subset U$.

there exists

1. An open subset $V \subset U$ such that V is closed under the action of S , and for which $V \cap \Delta_X^n = U'$.
2. A strong deformation retract $h_n : V \times I \rightarrow V$ onto U' such that for all $\sigma \in S$, $\sigma h_n(x, t) = h_n(\sigma x, t)$.

△

We are now ready for the main result of this section:

Theorem 2.18. *If X is symmetrically hyperlocally contractible, then $G^{(X)}$ is locally decomposably contractible.*

Proof. First, since $G^{(X)}$ is a topological group, we only need to show the statement around 0. So, assume U is an open neighbourhood of 0, and let E_U be the corresponding ensemble. The general method of the proof will be the following: we will use symmetric hyperlocal contractibility of X to get an open sub-ensemble $E_V \subset E_U$ such that $E_V(\emptyset) = \{*\}$ (i.e. V contains 0) and $E_V(g_1, \dots, g_n) = \emptyset$ whenever $\sum_{i=1}^n g_i \neq 0$. We define $V_n := G_n^{(X)} \cap V$. On each n -truncation $E_{V,n}$ of E_V , we will then construct a homotopy $h'_n := E_{V,n} \times I \rightarrow E_{V,n}$, which is the identity on each $E_{V,n}(g_1, \dots, g_k)$ for $k < n$, and for which the corresponding homotopy $V_n \times I \rightarrow V_n$ is a deformation retract to V_{n-1} .

By Lemma 2.16, this homotopy naturally extends to an $h_n : E_{V,n} \times I \rightarrow E_{V,n}$. We define $r_{n,n} : E_{V,n} \rightarrow E_{V,n}$ to be the identity, and $r_{n,n-1} : E_{V,n} \rightarrow E_{V,n-1}$ to be the map sending x to $h_n(x, 1)$. We then inductively define $r_{n,k}$ with $k < n$ to be $r_{k+1,k} \circ r_{n,k+1}$. We will then define the contraction $h : E_V \times I \rightarrow E_V$ on E_V to be

$$h(x, t) := \begin{cases} x & \text{if } t \in [0, \frac{1}{2^{n-1}}], x \in E_V(g_1, \dots, g_n) \\ h_k(r_{n,k}(x), 2^{k-1}t - 1) & \text{if } t \in (\frac{1}{2^{k-1}}, \frac{1}{2^{k-2}}], x \in E_V(g_1, \dots, g_n), 2 \leq k \leq n \end{cases}$$

which we can check is well defined and continuous. This descends to a contraction on V , as for all $x \in E_V(g_1, \dots, g_n)$, $h(x, 1) = r_{n,1}(x)$, which represents a point in V_1 . The only point in V_1 is 0.

Now we just have to construct V as well as the maps h_n . We do this inductively: $E_V(0)$ we just define to be X . For $n > 0$, we break things down into two cases: Assume first that $(g_1, \dots, g_n)$ has $g_i = 0$ for some i . Then without loss of generality, we assume $i = 1$, and then we define $E_V(g_1, \dots, g_n) :=$

$X \times E_V(g_2, \dots, g_n)$. The homotopy h_n on this we just define to be the constant homotopy. Now assume instead that $(g_1, \dots, g_n)$ has no zero in it. If $\sum_{i=1}^n g_i \neq 0$, then we define $E_V(g_1, \dots, g_n)$ to be the empty set. Otherwise, we do the following: In order to not get conflicting definitions, we fix an ordering on G (we can do this by the axiom of choice) and assume $g_1 \leq g_2 \leq \dots \leq g_n$. We will extend whatever we define here to all permutations via the rules $\sigma(E_V(g_1, \dots, g_n)) = E_V(g_{\sigma(1)}, \dots, g_{\sigma(n)})$. Of course, we still have to make sure our definitions of $E_V(g_1, \dots, g_n)$ and h_n on it respect the symmetries that preserve the ordering $(g_1, \dots, g_n)$. So, let $S \subset \Sigma_n$ be the subgroup that fixes $(g_1, \dots, g_n)$. Then we have the following data to work with:

1. A subgroup $S \subset \Sigma_n$.
2. An open subset $E_U(g_1, \dots, g_n) \subset X^n$.
3. An open subset $U' \subset \Delta_X^n$ defined as the set of all elements in $E_U(g_1, \dots, g_n)$ that represent the same formal sum as an element in some $E_V(g'_1, \dots, g'_{n-1})$. In practice, by Lemma 2.15 we can assume that one of the g'_i s is a sum of two g_j s, and the remaining g'_i s and g_j s are in one-to-one correspondence with each other. We can check that U' is open in Δ_X^n by noting that for each $1 \leq i < j \leq n$, if $X_{ij} = \{(x_1, \dots, x_n) \in X^n \mid x_i = x_j\}$, then $U' \cap X_{ij}$ is open in X_{ij} .

What we need to construct from this is

1. An open subset $E_V(g_1, \dots, g_n) \subset E_U(g_1, \dots, g_n)$ that is closed under the action of S , and for which $E_V(g_1, \dots, g_n) \cap \Delta_X^n = U'$.
2. A strong deformation retract $h_n : E_V(g_1, \dots, g_n) \times I \rightarrow E_V(g_1, \dots, g_n)$ onto U' such that for all $\sigma \in S$, $\sigma h_n(x, t) = h_n(\sigma x, t)$.

But this is precisely the condition of being symmetrically hyperlocally contractible, so we are done. This completes the proof. $\square$

Now we just need to show that a sufficiently large class of spaces X is symmetrically hyperlocally contractible. That sufficiently large class of spaces turns out to be the geometric realizations of simplicial sets, as we will show in the next section.

2.4 Geometric Realizations of Simplicial Sets are Hyperlocally Contractible

In the previous section we have shown that when X is hyperlocally contractible, then $G^{(X)}$ is locally decomposably contractible with respect to its defining decomposition structures. We will now show that all geometric realizations of simplicial sets are hyperlocally contractible, which will then prove that H_*^{conf} is a homology theory.

Theorem 2.19. *Every Geometric Realization of a Simplicial Set is Symmetrically Hyperlocally Contractible.*

Proof. Let A be a simplicial set. Since we are working within the category of k -spaces (note: this is the only place in the entire proof we use this fact), we have that $|A^n| = |A|^n$. This in particular means that the fat diagonal $\Delta_{|A|}^n$ of $|A|^n$ is a sub-simplicial set.

We will now define an open neighbourhood W of $\Delta_{|A|}^n$. First, we take a barycentric subdivision of A^n , and then we define W on each of the simplices in this subdivision. Note that when performing barycentric subdivision on an m -simplex, each m -simplex in the subdivision will have exactly one k -face that is contained in a k -face of the original simplex for each $k \leq m$. This in particular means that the intersection of an m -simplex in $B(A^n)$ and the set $\Delta_{|A|}^n$ is either empty or a k -simplex for some $k \leq m$. On simplices (in the subdivision) that do not intersect with $\Delta_{|A|}^n$, we define W to be empty, on all other simplices $[v_0, \dots, v_m]$ we define

$$W \cap [v_0, \dots, v_m] := \left\{ \sum_{i=0}^m t_i v_i \in [v_0, \dots, v_m] \mid \sum_{v_i \notin \Delta_{|A|}^n} t_i < \frac{1}{2} \right\}.$$

These patch together to define the open set W . We define on the same simplices a map $h : (W \cap [v_0, \dots, v_m]) \times I \rightarrow W \cap [v_0, \dots, v_m]$ as

$$h\left(\sum_{i=0}^m t_i v_i, t\right) = (1-t) \sum_{i=0}^m t_i v_i + \frac{t}{\sum_{v_i \in \Delta_{|A|}^n} t_i} \sum_{v_i \in \Delta_{|A|}^n} t_i v_i.$$

This is well defined, and these maps patch together to give a deformation retract $h : W \times I \rightarrow W$ onto the fat diagonal. The set W and the deformation retract are both symmetrically defined, so they both respect the action of Σ_n on $|A|^n$.

We can now use this to show Symmetric Hyperlocal Contractibility: Suppose we have the data of

1. A subgroup $S \subset \Sigma_n$
2. An open subset $U \subset X^n$.
3. An open subset $U' \subset \Delta_X^n$ of the fat diagonal that is closed under the action of S , and for which we have $U' \subset U$.

From this we need to construct

1. An open subset $V \subset U$ such that V is closed under the action of S , and for which $V \cap \Delta_X^n = U'$.
2. A strong deformation retract $h_n : V \times I \rightarrow V$ onto U' such that for all $\sigma \in S$, $\sigma h_n(x, t) = h_n(\sigma x, t)$.

Without loss of generality, we may assume $U \subset W$. We then construct V and h_n the following way: for any $x \in U'$, we can find an open $U_x \ni x$ and a $t \in [0, 1)$ such that $U_x \subset U$, $U_x \cap \Delta_X^n \subset U'$ and $h(U_x, [t, 1]) \subset U$. Note that the way we defined h tells us two things: $h(x, [t, 1]) = h(h(x, t), [0, 1])$ as sets, and $h(U_x, [t, 1])$ is open. This means in particular that we can replace U_x with $h(U_x, [t, 1])$ to get an open set for which $h(U_x, I) \subset U_x \subset U$ and $U_x \cap \Delta_X^n \subset U'$ holds. Then we define V as

$$V := \bigcap_{\sigma \in S} \sigma \left(\bigcup_{x \in U'} U_x \right)$$

and h_n as $h|_{V \times I}$. This has all the properties we want, which completes the proof. $\square$

This shows that $H_n^{\text{conf}}(X; G)$ is isomorphic to $H_n^{\text{sing}}(X; G)$ on all geometric realizations of simplicial sets. Since every CW complex is homotopy equivalent to such a space, this means the statement also holds for all CW complexes, as desired.

3 The Dold-Thom Theorem

In this section we will prove, using configuration homology, the Dold-Thom Theorem. For this, we will start with a definition, and then the statement of the theorem:

Definition 3.1 (Infinite Symmetric Product). If (X, b) is a based topological space, then its *infinite symmetric product* $\text{Sp}(X)$ is the inverse limit of the chain

$$X \rightarrow X^2/\Sigma_2 \rightarrow X^3/\Sigma_3 \rightarrow \dots$$

of based topological spaces, where Σ_n is the n th symmetric group acting on X^n by permuting coordinates, and the map $X^n/\Sigma_n \rightarrow X^{n+1}/\Sigma_{n+1}$ sends $[x_1, \dots, x_n]$ to $[x_1, \dots, x_n, b]$. $\triangle$

Theorem 3.2 (The Dold-Thom Theorem). *The natural functors $\tilde{H}_n^{DT}$ sending each based topological space X to $\pi_n(\text{Sp}(X))$ form a reduced homology theory on connected CW-complexes, with relative homology groups given by relative homotopy groups. [3]*

We will prove this by constructing a natural isomorphism of functors $\tilde{H}_n^{DT} \rightarrow \tilde{H}_n^{\text{conf}}$ (where H_*^{conf} has $\mathbb{Z}$ coefficients unless specified otherwise) for each n (in a way that commutes with the boundary maps in the long exact sequences) which, since we know Configuration Homology is a Homology Theory, implies the Dold-Thom theorem. To do this, we will first define the notion of a reduced formal sum space

Definition 3.3 (Reduced Formal Sum Space). Given a topological space X and a discrete abelian group G , the *reduced formal sum space* of X over G is the space

$$G_0^{(X)} := \left\{ \sum_{i=1}^n g_i x_i \in G^{(X)} \mid \sum_{i=1}^n g_i = 0 \right\}.$$

The corresponding reduced pre-sum space is denoted $G_0(X)$, and the decomposition structure on $G_0^{(X)}$ is denoted $q_0 : G_0(X) \rightarrow G_0^{(X)}$. $\triangle$

It is clear that we can define reduced configuration homology in the same way as the usual non-reduced configuration homology, just using reduced formal sum spaces instead of non-reduced ones.

We now have the following situation: both $\tilde{H}_*^{DT}$ and $\tilde{H}_*^{\text{conf}}$ are defined by taking a functor from based spaces to based spaces (Sp for the former, $\mathbb{Z}_0^{(-)}$ for the latter), and then applying π_* to it. The long exact sequence in both cases is given by the long exact sequence of relative homotopy groups. Therefore, a natural transformation $\text{Sp} \rightarrow \mathbb{Z}_0^{(-)}$ that is a weak homotopy equivalence for every space would give a natural isomorphism $\tilde{H}_*^{DT} \rightarrow \tilde{H}_*^{\text{conf}}$ of homology theories as desired.

On each based space (X, b) , we can construct a map $\text{Sp}(X) \rightarrow \mathbb{Z}_0^{(X)}$ via the rule $[x_1, \dots, x_n] \mapsto \sum_{i=1}^n (x_i - b)$. This will give our natural transformation.

Of course, we have to show that this is continuous and a weak equivalence, but the method we will use to show this generalizes to a far larger class of maps called Grothendieck Completion maps, which we will examine in the next section.

3.1 The Grothendieck Completion Map

We start with a definition

Definition 3.4 (Grothendieck Completion). Let $(M, +)$ be a commutative topological monoid. Its *Grothendieck Completion* is the topological group $\overline{M} := M \times M / \sim$ where $(a, b) \sim (c, d)$ if and only if there exists an $m \in M$ such that $a + d + m = b + c + m$. The group structure is given by pointwise addition, and the inverse is given by switching the two coordinates. The map $M \rightarrow \overline{M}, m \mapsto [(m, 0)]$ is called the *Grothendieck Completion Map*, and the map $M \times M \rightarrow \overline{M}, (a, b) \mapsto [(a, b)]$ is called the natural decomposition structure on $\overline{M}$. $\triangle$

Note that this is well defined: $\sim$ is an equivalence relation, as it is trivially reflexive and symmetric, and as for transitivity, we have that $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$ implies there exists $m_1, m_2 \in M$ such that

$$a + d + m_1 = b + c + m_1$$

$$c + f + m_2 = d + e + m_2$$

which means

$$a + f + (m_1 + m_2 + c + d) = b + e + (m_1 + m_2 + c + d),$$

which implies $(a, b) \sim (e, f)$ as desired. The fact that the group structure on $\overline{M}$ is well defined can also be checked manually.

One way to interpret the Grothendieck Completion is that it is the “freest” way to turn a commutative monoid into an abelian group. In other words, it can be defined via the universal property that all continuous monoid-homomorphisms $f : M \rightarrow G$ into a topological abelian group factor through $\overline{M}$ uniquely via the Grothendieck completion map. This is because any such f induces a map $M \times M \rightarrow G, (a, b) \rightarrow f(a) - f(b)$, and if $(a, b) \sim (c, d)$, then for some $m \in M$,

$$f(a) + f(d) + f(m) = f(b) + f(c) + f(m)$$

which implies $f(a) - f(b) = f(c) - f(d)$. This means the map $M \times M \rightarrow G$ descends to a map $\overline{M} \rightarrow G$. Precomposing this by the Grothendieck completion map gives us f , which proves f factors through $\overline{M}$, and does so uniquely because $\overline{M}$ is generated as a group by elements of the form $[(a, 0)]$.

Now, our goal for this section is to show that for a sufficiently nice class of topological monoids, the Grothendieck Completion map is a weak equivalence. For this, we first need two definitions:

Definition 3.5 (Decomposative Monoid). An abelian topological monoid M is called *decomposative* if $\overline{M}$ is locally decomposably contractible with respect to the natural decomposition structure $M \times M \rightarrow \overline{M}$. $\triangle$

Definition 3.6 (Cancellative Monoid). A commutative monoid M is called *cancellative* if for all $a, b, m \in M$, $a + m = b + m$ implies $a = b$. $\triangle$

Note that a monoid is cancellative if and only if its Grothendieck completion map is injective (the proof is a matter of unwinding definitions). Now we are ready to state the main theorem of this section:

Theorem 3.7. *Let M be a commutative path-connected topological monoid that is both cancellative and decomposative. Then its Grothendieck completion map is a weak equivalence.*

Proof. We have to show that for all n , the induced map $f_* : \pi_n(M) \rightarrow \pi_n(\overline{M})$ is an isomorphism.

First, we will show that it is injective. Let $\alpha : S^n \rightarrow M$ represent an element in $\pi_n(M)$, and suppose f_* sends it to 0. Since M is cancellative, we can treat it as a subset of $\overline{M}$, which means the condition that $f_*([\alpha]) = 0$ translates to there being a map $\beta : D^{n+1} \rightarrow \overline{M}$ that is α on the boundary. Now, this map is decomposable on the boundary, as $\alpha' : S^n \rightarrow M \times M, x \mapsto (\alpha(x), 0)$ is a decomposition of α . Therefore, we can use Theorem 2.12 to homotope β relative to the boundary to a decomposable map. For simplicity, assume β is already decomposable without loss of generality. Let $\beta' : D^{n+1} \rightarrow M \times M$ be its decomposition. This can be seen as a pair of maps $\beta'_1, \beta'_2 : D^{n+1} \rightarrow M$ with the

property that on ∂D^{n+1} , $\beta'_1(x) = \alpha(x) + \beta'_2(x)$ (by cancellativity). Treat the first map as having the domain $[0, 1] \times S^n / \{0\} \times S^n$ and the second map as having the domain $[1, 2] \times S^n / \{2\} \times S^n$ (in both cases, ∂D^{n+1} translates to $\{1\} \times S^n$). Also, since M is path-connected, we can assume without loss of generality that $\beta'_2([(2, x)]) = 0$ (as we can just homotope relative to ∂D^{n+1} to such a map). We can now write up a homotopy $H_1 : ([0, 1] \times S^n / \{0\} \times S^n) \times I \rightarrow M$ and a homotopy $H_2 : ([1, 2] \times S^n / \{2\} \times S^n) \times I \rightarrow M$ defined as

$$H_1((s, x), t) := \begin{cases} \beta'_1((1+t)s, x) & \text{if } (1+t)s \leq 1 \\ \alpha(x) + \beta'_2((1+t)s, x) & \text{otherwise.} \end{cases}$$

$$H_2((s, x), t) := \beta'_2(\min((1+t)s, 2), x)$$

The first of these homotopies is continuous by the pasting lemma (as $\beta'_1(1, x) + \alpha(x) + \beta'_2(1, x)$), and the second is trivially continuous. Applying these two homotopies simultaneously gives a homotopy of β' , which gives a homotopy of β (for an illustration of this homotopy, see 7). However, note that since $H_1((1, x), t) = \alpha(x) + H_2((1, x), t)$ for all t , the induced homotopy on β is actually relative to ∂D^{n+1} . With this homotopy, we can exchange β for another null-homotopy of α which has a decomposition $D^{n+1} \rightarrow M \times M$ that is the constant $H_2((s, x), 1) = \beta'_2([(2, x)]) = 0$ on the second coordinate. This means α can be homotoped to 0 in M , which means $[\alpha] = 0 \in \pi_n(M)$ as desired.

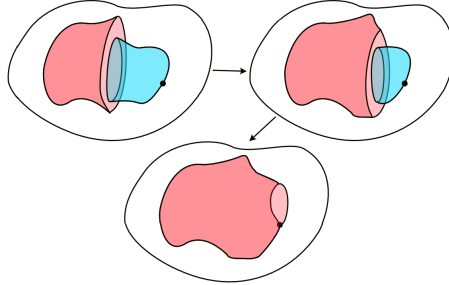


Figure 7: An illustration of the homotopies H_1 and H_2 in the proof of Theorem 3.1. The red and blue sections represent $\beta_1 : D^2 \rightarrow M$ and $\beta_2 : D^2 \rightarrow M$ respectively in the beginning, middle, and end of the homotopy. The black dot represents $0 \in M$.

The proof of surjectivity is nearly identical: Reusing variables, let $\alpha : D^n \rightarrow \overline{M}$ be a map representing an element of $\pi_n(\overline{M})$. By Theorem 2.12, we may choose this α to have a decomposition $\alpha' : D^n \rightarrow M \times M$. From that point, the proof is identical to the one for injectivity: we homotope α' (in such a way that it descends to a homotopy of α relative to ∂D^n) to a map which is 0 on the second coordinate. If we call the resulting map into the first coordinate $\beta : D^n \rightarrow M$, then we know that $[\alpha] = f_*([\beta])$, which proves surjectivity.

This completes the proof. $\square$

Now all we have to do is apply this to the Dold-Thom Theorem.

3.2 The Generalized Dold-Thom Theorem

In this section, we will prove not just the Dold-Thom theorem, but a generalized version of it. We can define, for any discrete commutative monoid M and topological space X , the formal sum space $M^{(X)}$ identically to how it is defined for groups. We can also define, for any based space X , the space $M_r^{(X)} := M^{(X)} / \sim$ where two formal sums are equivalent if they agree everywhere except potentially on the basepoint. We refer to $M_r^{(X)}$ as a basepoint-reduced formal sum space. Note that if $M = G$ is a group, then $G_r^{(X)}$ is naturally homeomorphic to $G_0^{(X)}$ via the map $G_r^{(X)} \rightarrow G_0^{(X)}, [\sum_{i=1}^n g_i x_i] \mapsto \sum_i^n (g_i x_i - g_i b)$.

This is relevant to the Dold-Thom theorem, as we can show that $\mathrm{Sp}(X)$ is homeomorphic to $\mathbb{N}_r^{(X)}$ via the map $[x_1, \dots, x_n] \rightarrow [\sum_{i=1}^n x_i]$ for all based spaces X (to see that this is well-defined and a homeomorphism is a matter of unwinding definitions). This allows us to state the main theorem of this section:

Theorem 3.8 (Generalized Dold-Thom Theorem). *Let M be a discrete commutative cancellative monoid. Then the functors H_n^{DTM} which send a based topological space X to $\pi_n(M_r^{(X)})$ form a reduced homology theory on connected CW-complexes, with the long exact sequence in homology given by the long exact sequence of relative homotopy groups. The coefficient of this homology theory is $\overline{M}$.*

We will prove this using two lemmas. First, we have

Lemma 3.9. *For any discrete commutative monoid M and based space X , the induced map $M_r^{(X)} \rightarrow \overline{M}_r^{(X)}$ (or equivalently, the induced map $M_r^{(X)} \rightarrow \overline{M}_0^{(X)}$) is a Grothendieck Completion map (in the sense that it has the necessary universal property).*

Proof. We have to show that the induced map $\overline{M}_r^{(X)} \rightarrow \overline{M}_r^{(X)}$ is a homeomorphism. We will explicitly construct its inverse by the following rule: An element $[\sum_{i=1}^n [(a_i, b_i)] x_i] \in \overline{M}_r^{(X)}$ maps to $[(\sum_{i=1}^n a_i x_i), [\sum_{i=1}^n b_i x_i]]$. If this is well-defined, then it is a valid inverse, and it is continuous by the fundamental property of quotients, so we just need to check that it is well-defined. We only need to check this on generators, as our map is a homomorphism, and on a generator of the form $[a, b]x$, our map is well-defined by unwinding definitions. This completes the proof. $\square$

For the second lemma, we need to define a slightly stronger version of symmetric hyperlocal contractibility for based spaces, which we will call basepoint-wise symmetric hyperlocal contractibility. Essentially, this definition is identical to definition 2.17, except that we require the following property to hold for the map $h_n : V \times I \rightarrow V$: if $h_n(x_1, \dots, x_n, t) = (x'_1, \dots, x'_n)$, then $x_i = b$ should imply $x'_i = b$. In other words, h_n coordinate-wise preserves basepoints. Note

that the proof of Theorem 2.19 also works to show every geometric realization of a simplicial set is a basepoint-wise symmetrically hyperlocally contractible space, as long as we assume the basepoint is a vertex (which we can always do). Now we can state the lemma:

Lemma 3.10. *If M is a discrete commutative cancellative monoid, and X is a basepoint-wise symmetrically hyperlocally contractible space, then $M_r^{(X)}$ is a decomposative monoid.*

Proof. By Lemma 3.9, we have to show that $\overline{M}_0^{(X)}$ is locally decomposably contractible with respect to the decomposition structure $j : M_r^{(X)} \times M_r^{(X)} \rightarrow \overline{M}_0^{(X)}$. We will do this by finding an open neighbourhood $U \subset \overline{M}_0^{(X)}$ of 0 for which there exists a pair of homotopies $r : U \times I \rightarrow U$ and $r' : j^{-1}(U) \times I \rightarrow U$ such that r is a contraction (to 0) and the diagram

$$\begin{array}{ccc} j^{-1}(U) \times I & \xrightarrow{r'} & j^{-1}(U) \\ \downarrow j \times \text{id} & & \downarrow j \\ U \times I & \xrightarrow{r} & U \end{array}$$

commutes. In the proof of Theorem 2.18, we constructed the analogous maps for the decomposition structure $q_0 : \overline{M}_0(X) \rightarrow \overline{M}_0^{(X)}$. For explaining the rest of the proof, we will use the diagram

$$\begin{array}{ccccc} & & j^{-1}(U) \times I & \xrightarrow{r'} & j^{-1}(U) \\ & \nearrow & & \searrow & \nearrow \\ k^{-1}(U) \times I & \xrightarrow{l'} & k^{-1}(U) & & U \times I \xrightarrow{r} U \\ & \searrow & & \nearrow & \searrow \\ & & q_0^{-1}(U) \times I & \xrightarrow{l} & q_0^{-1}(U) \end{array}$$

as visual aide. For the rest of this proof, we will refer to this as “the big diagram”. Our goal in the end is to construct r and r' . The subspace $U \subset \overline{M}_0^{(X)}$ and the maps r and l are constructed as in the proof of Theorem 2.18. Because we know X is actually basepoint-wise symmetrically hyperlocally contractible, we can construct l in such a way that it coordinate-wise preserves basepoints (by just following the construction in the proof of Theorem 2.18, but assuming each h_n in the proof coordinate-wise preserves basepoints), so we will assume this holds. Now, we will define

$$M(X) := \prod_{i=1}^{\infty} (M \times X)^i.$$

And we will define $M_0(X)$ to be the subspace of $M(X) \times M(X)$ on which the coefficients in the first coordinate and the coefficients in the second coordinate sum to the same element of M . There is an induced map $f : M_0(X) \rightarrow$

$\overline{M}_0(X)$, $((m_i, x_i)_{i=1}^n, (m_i, x_i)_{i=n+1}^{n'}) \mapsto (s_i \overline{m_i}, x_i)_{i=1}^{n'}$, where $s_i = 1$ if $i \leq n$ and -1 otherwise. This is a homeomorphism on connected components (i.e. it is a homeomorphism when restricting both the domain and the codomain to connected components that are compatible with respect to f). Composing f with q defines $k : M_0(X) \rightarrow \overline{M}_0^{(X)}$, and l induces the map l' in the big diagram.

Next, we will define a map $u : M_0(X) \rightarrow M_r^{(X)} \times M_r^{(X)}$ via the rule

$$u((m_i, x_i)_{i=1}^n, (m'_i, x'_i)_{i=1}^{n'}) := ([\sum_{i=1}^n m_i x_i], [\sum_{i=1}^{n'} m'_i x'_i]).$$

The map u is continuous by the definition of $M_r^{(X)}$, and surjective, as

$$\begin{aligned} ([\sum_{i=1}^n m_i x_i], [\sum_{i=1}^{n'} m'_i x'_i]) &= ([\sum_{i=1}^n m_i x_i + \sum_{i=1}^{n'} m'_i b], [\sum_{i=1}^{n'} m'_i x'_i + \sum_{i=1}^n m_i b]) \\ &= u((m_i, x_i)_{i=1}^n * (m'_i, b)_{i=1}^{n'}, (m'_i, x'_i)_{i=1}^{n'} * (m_i, b)_{i=1}^n) \end{aligned}$$

where $*$ stands for concatenation. We also know that u is a quotient map: by definition a set $V \subset M_r^{(X)} \times M_r^{(X)}$ is open if and only if its preimage is open in $M(X) \times M(X)$ with respect to the defining quotient map $u' : M(X) \times M(X) \rightarrow M_r^{(X)} \times M_r^{(X)}$. Now, u is just a restriction of u' to $M_0(X)$, so we have to show that $u'^{-1}(V)$ is open if and only if $u'^{-1}(U) \cap M_0(X)$ is open. Rightward implication is clear, and leftward implication follows from being able to construct a map $M(X) \times M(X) \rightarrow M_0(X)$ via the rule

$$((m_i, x_i)_{i=1}^n, (m'_i, x'_i)_{i=1}^{n'}) \mapsto ((m_i, x_i)_{i=1}^n * (m'_i, b)_{i=1}^{n'}, (m'_i, x'_i)_{i=1}^{n'} * (m_i, b)_{i=1}^n)$$

where $*$ again stands for concatenation. This map preserves the quotient relations, which means pulling back via this map gets us from $u'^{-1}(V) \cap M_0(X)$ to $u'^{-1}(V)$, which means if the former is open, then so is the latter, as desired. This shows u is a quotient map.

The final step of this proof is to argue that u induces the map r' in the big diagram. If r' is induced as a function (i.e. there is a well defined function that makes the diagram commute), then it is continuous by u being a quotient map. So, we just have to check that r' is well defined. The short version of how we can check this is the following: the equivalence relation $\sim$ on $M_0(X)$ that defines the quotient map $M_0(X) \rightarrow M_r^{(X)} \times M_r^{(X)}$ is generated by

1. Permutations on either side: For any $\sigma \in \Sigma_n$, $((m_i, x_i)_{i=1}^n, (m'_i, x'_i)_{i=1}^{n'}) \sim ((m_{\sigma(i)}, x_{\sigma(i)})_{i=1}^n, (m'_i, x'_i)_{i=1}^{n'})$, and same on the other side.
2. Addition on either side: if $x_n = x_{n+1}$, then $((m_1, x_1, \dots, m_{n-1}, x_{n-1}, m_n + m_{n+1}, x_n), (m'_i, x'_i)_{i=1}^{n'}) \sim ((m_i, x_i)_{i=1}^{n+1}, (m'_i, x'_i)_{i=1}^{n'})$, and same on the other side.

3. Appending a 0 either side: if $m_{n+1} = 0$, then $((m_i, x_i)_{i=1}^n, (m'_i, x'_i)_{i=1}^{n'}) \sim ((m_i, x_i)_{i=1}^{n+1}, (m'_i, x'_i)_{i=1}^{n'})$, and same on the other side.
4. Appending a basepoint on both sides: if $m_{n+1} = m'_{n'+1}$ and $x_{n+1} = x'_{n'+1} = b$, then $((m_i, x_i)_{i=1}^n, (m'_i, x'_i)_{i=1}^{n'}) \sim ((m_i, x_i)_{i=1}^{n+1}, (m'_i, x'_i)_{i=1}^{n'+1})$. This time we have to do it simultaneously on both sides, as the m_i coefficients have to have the same sum as the m'_i ones.

The first three of these relations is preserved by construction (see the proof of Theorem 2.18), and the final one is preserved by l' coordinate-wise preserving the basepoints. This means r' is indeed well defined as desired, which completes the proof. $\square$

With this, we have proven the Generalized Dold-Thom Theorem. In the next section, we examine an application of the Dold-Thom Theorem to the Hurewicz Map.

3.3 The Hurewicz Map

We start with a refresher of what the Hurewicz map is:

Definition 3.11 (The Hurewicz Map). If (X, b) is a based topological space, the Hurewicz map is, for each n , a function $\pi_n(X, b) \rightarrow H_n(X, b) =: \tilde{H}_n(X)$ defined by sending an element in $\pi_n(X, b)$ represented by $\alpha : \Delta^n \rightarrow X$ to $[\alpha] \in \tilde{H}_n^{\text{sing}}(X)$. $\triangle$

For $n \neq 0$, this is always a group homomorphism, and it is natural in the sense of category theory. In Section 2.1 we have constructed an explicit natural isomorphism $W_* : H_*^{\text{sing}} \rightarrow H_*^{\text{conf}}$. Composing this with the Hurewicz map $\pi_* \rightarrow \tilde{H}_*^{\text{sing}}$ gives a natural transformation $\pi_* \rightarrow \tilde{H}_*^{\text{conf}}$. In the spirit of category theory, we will also call this the Hurewicz map. This Hurewicz map is induced from the map $f : X \rightarrow \mathbb{Z}_0^{(X)}, x \mapsto x - b$. We can check this explicitly: given some $\alpha : \Delta^n \rightarrow X$ representing an element of $\pi_n(X)$, the map $f \circ \alpha$ can be decomposed as the formal sum of maps $\alpha - b$ (where b is the constant map). As seen in the proof of Theorem 2.13, $W_*([\alpha - b]) = [\alpha - b] \in \pi_n(\mathbb{Z}_0^{(X)})$, which is exactly what we need for f to induce the Hurewicz map.

Now, f factors through $\mathbb{N}_r^{(X)} \rightarrow \mathbb{Z}_r^{(X)}$ via the map $f' : X \rightarrow \mathbb{N}_r^{(X)}, x \mapsto [x]$. Since the map $\mathbb{N}_r^{(X)} \rightarrow \mathbb{Z}_r^{(X)}$ is a weak equivalence inducing an isomorphism $\tilde{H}_n^{\text{DT}}(X) \rightarrow \tilde{H}_n^{\text{conf}}(X)$, we can again in the spirit of category theory call $f'_* : \pi_n(X) \rightarrow \pi_n(\mathbb{N}_r^{(X)})$ the Hurewicz map.

We end this section by proving a powerful consequence of this description of the Hurewicz map:

Theorem 3.12. *Let M be a commutative topological monoid. Then the Hurewicz map $\pi_n(M) \rightarrow \tilde{H}_n(M)$ is injective.*

Proof. The Hurewicz map is induced by the map $M \rightarrow \mathbb{N}_r^{(M)}, m \mapsto [m]$. The map $\mathbb{N}_r^{(M)} \rightarrow M, \sum_{i=1}^n [m_i] \mapsto \sum_{i=1}^n m_i$ is a retract of this, so the Hurewicz map $\pi_n(M) \rightarrow \pi_n(\mathbb{N}_r^{(M)})$ must be injective. This completes the proof. $\square$

We will end this section by noting that there is a stronger result in the literature: A commutative topological monoid is always weakly equivalent to a product of Eilenberg-MacLane spaces. [4, p. 483]

4 Stable Homotopy

In this section, we will examine how configuration homology interacts with stability. For this, we first need to explain what we mean by stability: generally, a topological invariant on based spaces is stable if it is preserved by suspension, up to a shift of dimensions. In our case, those topological invariants will always have the form of a series of functors F_i with $i \in \mathbb{Z}$ from some subclass of based topological spaces to a category $\mathcal{C}$ (which will usually be just the category of abelian groups). Stability then will be expressed by, for all i , a natural isomorphism of functors $F_i \rightarrow F_{i+1}\Sigma$.

All reduced homology and cohomology theories are stable, and in both cases the natural isomorphisms witnessing stability are induced by the long exact sequence of (co)homology for the sequence $X \hookrightarrow CX \rightarrow \Sigma X$, where CX is the based cone of X (i.e. $X \times I / (\{b\} \times I \cup X \times \{0\})$). However, for many (co)homology theories, there is a more explicit description of the maps $\tilde{H}_n(X) \rightarrow \tilde{H}_{n+1}(\Sigma X)$ and $\tilde{H}^n(X) \rightarrow \tilde{H}^{n+1}(\Sigma X)$ respectively. For both singular and cellular homology, these maps witnessing stability are induced by chain maps $\tilde{C}_*^{\text{sing}}(X) \rightarrow \tilde{C}_{*+1}^{\text{sing}}(\Sigma X)$ and $\tilde{C}_*^{\text{cell}}(X) \rightarrow \tilde{C}_{*+1}^{\text{cell}}(\Sigma X)$ respectively, and similarly the corresponding map on cellular cohomology is induced by a chain map $\tilde{C}_{\text{cell}}^*(X) \rightarrow \tilde{C}_{\text{cell}}^{*+1}(\Sigma X)$. As we will see in this section, we can also find a natural description of stability for reduced configuration homology. In the process, we will explore spectra, which are the primary objects of Stable Homotopy Theory, and see how they can be used to represent both homology and cohomology theories. For a more comprehensive overview of Stable Homotopy Theory, see [1].

For the sake of convenience, in this section every (co)homology theory is reduced.

4.1 Stability of Configuration Homology

Our goal is to find an explicit isomorphism $\pi_k(G_0^{(X)}) \rightarrow \pi_{k+1}(G_0^{(\Sigma X)})$ that witnesses the stability of configuration homology.

For any based space Z , applying the functor Σ to $[S^k, Z]$ gives us a map $\varphi_k : \pi_k(Z) \rightarrow \pi_{k+1}(\Sigma Z)$, which is a group homomorphism for $k > 0$ (essentially by definition). Note that unlike for the analogous maps $\tilde{H}_k(Z) \rightarrow \tilde{H}_{k+1}(\Sigma Z)$ and $\tilde{H}^k(Z) \rightarrow \tilde{H}^{k+1}(\Sigma Z)$ for (co)homology groups, these do not need to be isomorphisms, as homotopy groups are not stable. Applying φ_k to the case

where $Z = G_0^{(X)}$, we get a homomorphism $\pi_k(G_0^{(X)}) \rightarrow \pi_{k+1}(\Sigma G_0^{(X)})$. To get to our desired isomorphism $\pi_k(G_0^{(X)}) \rightarrow \pi_{k+1}(G_0^{(\Sigma X)})$, we will compose this with the homomorphism $\pi_{k+1}(\Sigma G_0^{(X)}) \rightarrow \pi_{k+1}(G_0^{(\Sigma X)})$ induced from the map $\Sigma G_0^{(X)} \rightarrow G_0^{(\Sigma X)}, [(\sum_{i=1}^n g_i x_i, t)] \mapsto \sum_{i=1}^n g_i [(x_i, t)]$, which is continuous by the fundamental property of quotients. We can state this in a proper theorem:

Theorem 4.1. *The homomorphism $\pi_k(G_0^{(X)}) \rightarrow \pi_{k+1}(G_0^{(\Sigma X)})$ as described above is the isomorphism witnessing the stability of configuration homology.*

Proof. The stability of a homology theory arises from the long exact sequence associated to $X \hookrightarrow CX \rightarrow \Sigma X$. Specifically, $\tilde{H}_k^{\text{conf}}(X) \rightarrow \tilde{H}_{k+1}^{\text{conf}}(\Sigma X)$ should be the inverse of the boundary map $\partial_{k+1} : \tilde{H}_{k+1}^{\text{conf}}(\Sigma X) \rightarrow \tilde{H}_k^{\text{conf}}(X)$. This boundary map is induced by the boundary map of relative homotopy groups $\pi_{k+1}(G_0^{(CX)}, G_0^{(X)}) \rightarrow \pi_k(G_0^{(X)})$. Thinking of an element in $\pi_{k+1}(G_0^{(CX)}, G_0^{(X)})$ as a map $CS^k \rightarrow G_0^{(CX)}$ which is 0 on $\{b\} \times I$ and maps into $G_0^{(X)}$ on $S^k \times \{1\}$, we can explicitly write a map $\pi_k(G_0^{(X)}) \rightarrow \pi_{k+1}(G_0^{(CX)}, G_0^{(X)})$ which sends some $S^k \rightarrow G_0^{(X)}, s \mapsto \sum_{i=1}^{n_s} g_{i,s} x_{i,s}$ to $CS^k \rightarrow G_0^{(CX)}, (s, t) \mapsto \sum_{i=1}^{n_s} g_{i,s} [(x_{i,s}, t)]$ (this will be continuous by the fundamental property of quotients, and well defined as any homotopy on the former will induce a homotopy on the latter). We can directly confirm that composing this with ∂_{k+1} gives us the identity on $\pi_k(G_0^{(X)})$, which, since we already know ∂_{k+1} is an isomorphism, means our map $\pi_k(G_0^{(X)}) \rightarrow \pi_{k+1}(G_0^{(CX)}, G_0^{(X)})$ must be its inverse. Applying the isomorphism $\pi_{k+1}(G_0^{(CX)}, G_0^{(X)}) \rightarrow \pi_{k+1}(G_0^{(\Sigma X)})$, we get a map $\pi_k(G_0^{(X)}) \rightarrow \pi_{k+1}(G_0^{(\Sigma X)})$. It is a matter of unwinding definitions to show this is the same as the map we constructed earlier, so this completes the proof. $\square$

We will let the construction of first taking the suspension $\pi_k(G_0^{(X)}) \rightarrow \pi_{k+1}(\Sigma G_0^{(X)})$ and then applying a homomorphism $\pi_{k+1}(\Sigma G_0^{(X)}) \rightarrow \pi_{k+1}(G_0^{(\Sigma X)})$ induced from a map $\Sigma G_0^{(X)} \rightarrow G_0^{(\Sigma X)}$ serve as motivation for the following definitions:

Definition 4.2 (Spectrum). A *spectrum* (X_i, γ_i) comprises a sequence of based spaces X_i along with a sequence of maps called structure maps $\gamma_i : \Sigma X_i \rightarrow X_{i+1}$ for each natural number i . $\triangle$

Definition 4.3 (Homotopy Groups of a Spectrum). Given a spectrum $X = (X_i, \gamma_i)$, its k -th *homotopy group* (where k is an integer that may be negative) $\pi_k(X)$ is the colimit of the sequence

$$\pi_k(X_0) \rightarrow \pi_{k+1}(\Sigma X_0) \rightarrow \pi_{k+1}(X_1) \rightarrow \pi_{k+2}(\Sigma X_1) \rightarrow \dots$$

in the category of abelian groups. We may ignore the first $-2k+3$ terms in this sequence, as they are either undefined or are not an abelian group. $\triangle$

Definition 4.4 (Map of Spectra). Given a pair of spectra $X = (X_i, \gamma_i)$ and $X' = (X'_i, \gamma'_i)$, a *map of spectra* $f : X \rightarrow X'$ is a series of continuous functions $f_i : X_i \rightarrow X'_i$ such that for each i , $\gamma'_{i+1} \circ \Sigma f_i = f_{i+1} \circ \gamma_i$. $\triangle$

Any map of spectra $f : X \rightarrow X'$ induces a map on the homotopy groups $f_* : \pi_*(X) \rightarrow \pi_*(X')$, as the diagram

$$\begin{array}{ccccccc} \pi_k(X_0) & \longrightarrow & \pi_{k+1}(\Sigma X_0) & \longrightarrow & \pi_{k+1}(X_1) & \longrightarrow & \pi_{k+2}(\Sigma X_{1*}) \longrightarrow \dots \\ \downarrow f_{0*} & & \downarrow \Sigma f_{0*} & & \downarrow f_{1*} & & \downarrow \Sigma f_{1*} \\ \pi_k(X'_0) & \longrightarrow & \pi_{k+1}(\Sigma X'_0) & \longrightarrow & \pi_{k+1}(X'_1) & \longrightarrow & \pi_{k+2}(\Sigma X'_1) \longrightarrow \dots \end{array}$$

commutes. If this map is an isomorphism for all k , then f is called a weak equivalence.

One way to think of spectra is that they are “topological spaces modulo unstable behaviour”. For example, while homotopy groups are unstable in topological spaces, they are stable for spectra: given a spectrum (X_i, γ_i) , its suspension is the spectrum $(\Sigma X_i, \Sigma \gamma_i)$, and the suspension map for homotopy groups is induced by

$$\begin{array}{ccccccc} \pi_k(X_0) & \longrightarrow & \pi_{k+1}(\Sigma X_0) & \longrightarrow & \pi_{k+1}(X_1) & \longrightarrow & \pi_{k+2}(\Sigma X_1) \longrightarrow \dots \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \pi_{k+1}(\Sigma X_0) & \longrightarrow & \pi_{k+2}(\Sigma^2 X_0) & \longrightarrow & \pi_{k+2}(\Sigma X_1) & \longrightarrow & \pi_{k+3}(\Sigma^2 X_1) \longrightarrow \dots \end{array}$$

However, note that by composing pairs of maps on the top and bottom, we can compactify this diagram, without changing the colimit of either row, to

$$\begin{array}{ccccccc} \pi_k(X_0) & \longrightarrow & \pi_{k+1}(\Sigma X_0) & \longrightarrow & \pi_{k+2}(\Sigma X_1) & \longrightarrow & \pi_{k+3}(\Sigma X_2) \longrightarrow \dots \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \pi_{k+1}(\Sigma X_0) & \longrightarrow & \pi_{k+2}(\Sigma X_1) & \longrightarrow & \pi_{k+3}(\Sigma X_2) & \longrightarrow & \pi_{k+4}(\Sigma X_3) \longrightarrow \dots \end{array}$$

which makes it clear that the induced map on the colimits is an isomorphism. Following the “a spectrum is a space modulo unstable behaviour” line of thinking, we can construct a functor Σ^∞ called infinite suspension from based topological spaces to spectra, which sends each space X to the spectrum $\Sigma^\infty X := (\Sigma^i X, \text{id}_{\Sigma^{i+1} X})$. We can think of this as the “stabilization” of X . The groups $\pi_k(\Sigma^\infty X)$ are called the stable homotopy groups of X , and we will write them as $\pi_k^s(X)$.

Returning to the topic of configuration homology, we can define the following class of spectra:

Definition 4.5 (Formal Sum Spectrum). Given a spectrum $X := (X_i, \gamma_i)$ and a discrete abelian group G , we can define the *formal sum spectrum* $G_0^{(X)}$ of X over G to be $(G_0^{(X_i)}, \gamma'_i)$, where γ'_i is the composition $\Sigma G_0^{(X_i)} \rightarrow G_0^{(\Sigma X_i)} \rightarrow G_0^{(X_{i+1})}$, with the second map being induced by γ_i . $\triangle$

Given a spectrum X and an abelian group G , it is natural to ask what the homotopy groups of $G_0^{(X)}$ are. The notion of singular homology translates

naturally to the setting of spectra, as we can define $H_k^{\text{sing}}(X; G)$ as the colimit of

$$H_k^{\text{sing}}(X_0; G) \rightarrow H_{k+1}^{\text{sing}}(\Sigma X_0; G) \rightarrow H_{k+1}^{\text{sing}}(X_1; G) \rightarrow \dots$$

Applying the map W_* constructed in section 2.1 to this on every level gives an isomorphism $H_k^{\text{sing}}(X; G) \rightarrow \pi_k(G_0^{(X)})$, so, like in the case of topological spaces, there is a correspondence between the homology groups of a spectrum and the homotopy groups of the corresponding formal sum spectrum. If the structure maps $\Sigma X_i \rightarrow X_{i+1}$ induce an isomorphism on all homology groups, then the maps $H_k(X_i; G) \rightarrow H_{k+1}(X_{i+1}; G)$ are all isomorphisms, which in particular means that the homology of the spectrum is the same as the homology of X_0 . Taking the corresponding formal sum spectrum, we can infer that the maps $\pi_k(G_0^{(X_i)}) \rightarrow \pi_{k+1}(G_0^{(X_{i+1})})$ are isomorphisms, which means the homotopy groups of the formal sum spectrum are the same as the homotopy groups of $G_0^{(X_0)}$ (note: for this deduction to work, we use the fact that $G_0^{(X_i)}$ for $i > 0$ is path connected, as otherwise the spectrum may have some non-trivial negative homotopy groups).

We can examine this property in more detail: suppose Z is a spectrum for which $\pi_k(Z_i) \rightarrow \pi_{k+1}(Z_{i+1})$ is an isomorphism for all $k \geq 0$ (for $k = 0$ this means it is a bijection). This means the map $[S^k, Z_i] \rightarrow [S^{k+1}, Z_{i+1}] \xrightarrow{\sim} [S^k, \Omega Z_{i+1}]$ is an isomorphism, and since it is induced by the adjunction $Z_i \rightarrow \Omega Z_{i+1}$ of the structure map, we can conclude that $\pi_k(Z_i) \rightarrow \pi_{k+1}(Z_{i+1})$ is an isomorphism for all $k \geq 0$ if and only if $Z_i \rightarrow \Omega Z_{i+1}$ is a weak equivalence. This motivates the following definition:

Definition 4.6 (Ω -spectrum, -1 -connected). A spectrum $X = (X_i, \gamma_i)$ is called an Ω -spectrum if the adjunctions $X_i \rightarrow \Omega X_{i+1}$ of the structure maps are weak equivalences. If on top of that, each X_i with $i > 0$ is path-connected, then X is said to be a -1 -connected Ω -spectrum. $\triangle$

An important special case of this is the following:

Definition 4.7 (Eilenberg-MacLane spectrum). A spectrum X is called an *Eilenberg-MacLane spectrum* if it is an Ω -spectrum, and each X_i is a $K(G, i)$ space. Equivalently, if X is a -1 -connected Ω -spectrum such that X_0 is a $K(G, 0)$ space. $\triangle$

We have now proven the following statements:

Lemma 4.8. *If X is an Ω -spectrum, then the natural map $\pi_k(X_0) \rightarrow \pi_k(X)$ is an isomorphism for $k \geq 0$. If X is also -1 -connected, then $\pi_k(X) = 0$ for $k < 0$.*

Lemma 4.9. *If X is a spectrum such that each structure map induces an isomorphism on homology groups, then $G_0^{(X)}$ is a -1 -connected Ω -spectrum.*

These lemmas have the following important corollaries for our purposes:

Corollary 4.10. *The spectrum $G_0^{(\Sigma^\infty X)}$ is a -1 -connected Ω -spectrum.*

Corollary 4.11. *The spectrum $G_0^{(\Sigma^\infty S^0)}$ is an Eilenberg-MacLane spectrum.*

We have now done enough groundwork to explain how spectra represent homology and cohomology theories.

4.2 Spectra Induce Homology Theories

A modern approach to homology and cohomology involves representing (co)homology theories with spectra. In this section we will show how spectra represent homology theories, and we will find an explicit spectrum that represents configuration homology. In the next section, we will discuss how spectra represent cohomology theories, which we will then use to define configuration cohomology.

We will now show how a spectrum induces a homology theory. Given a based space X and a spectrum $Y = (Y_i, \gamma_i)$, we can define the spectrum $X \wedge Y := (X \wedge Y_i, \text{id}_X \wedge \gamma_i)$ (note that this definition works because $\Sigma(X \wedge Y_i)$ is naturally homeomorphic to $X \wedge \Sigma Y_i$). The main goal of this section will be to prove that for any spectrum Y , the functors H_k^Y defined as sending each based space X to $\pi_k(X \wedge Y)$ form a reduced homology theory on CW-complexes, with the long exact sequence of homology being induced by the long exact sequence of relative homotopy groups.

Before we proceed with the proof, we will note a consequence of it: In the special case that $Y = G_0^{(\Sigma^\infty S^0)}$, we actually have a natural map of spectra $X \wedge G_0^{(\Sigma^\infty S^0)} \rightarrow G_0^{(\Sigma^\infty X)}$. This is defined, on each level $X \wedge G_0^{(S^i)} \rightarrow G_0^{(X \wedge S^i)}$, as sending $[(x, \sum_{j=1}^n g_j s_j)]$ to $\sum_{j=1}^n g_j [(x, s_j)]$. We can confirm that this is continuous by the fundamental property of quotients, and that it commutes with the structure maps. Since $G_0^{(\Sigma^\infty X)}$ is a -1 -connected Ω -spectrum, its homotopy groups are the same as the homotopy groups of $G_0^{(X)}$, which means we have a natural transformation of homology theories $H_*^{G_0^{(\Sigma^\infty S^0)}} \rightarrow \tilde{H}_*^{\text{conf}}$. Furthermore, this natural transformation is an isomorphism on S^0 , which means it has to be an isomorphism on every CW-complex. This reasoning goes both ways: if we could prove without relying on singular homology that $G_0^{(\Sigma^\infty X)}$ is an Ω -spectrum (showing that it is also -1 -connected is not necessary, as we already know that a reduced formal sum space over a path connected space is path connected) and that $X \wedge G_0^{(\Sigma^\infty S^0)} \rightarrow G_0^{(\Sigma^\infty X)}$ is a weak equivalence of spectra for all X , that would imply that configuration homology is a homology theory. We will not give such a proof in this thesis, but it is an avenue we believe to be worth exploring.

We will now show that H_*^Y gives a homology theory for any spectrum Y : We start by showing this is the case wherever Y is an infinite suspension. Reusing the variable Y , we will write it as the spectrum $\Sigma^\infty Y$. The only Eilenberg-Steenrod axiom that does not immediately hold is excision, so we will focus on that: we have to show that, given $A \subset B \subset X$ with the basepoint $b \in B - A$, if the closure of A is in the interior of B , then

$$\pi_k((X - A) \wedge \Sigma^\infty Y, (B - A) \wedge \Sigma^\infty Y) \rightarrow \pi_k(X \wedge \Sigma^\infty Y, B \wedge \Sigma^\infty Y)$$

is an isomorphism. We only need to show that, for any given k , for sufficiently large i , the map

$$\pi_{k+i}((X - A) \wedge \Sigma^i Y, (B - A) \wedge \Sigma^i Y) \rightarrow \pi_{k+i}(X \wedge \Sigma^i Y, B \wedge \Sigma^i Y)$$

is an isomorphism. To do this, we will use the homotopy excision theorem [2, p. 133], which states the following: given three spaces $A' \subset B' \subset X'$ with a basepoint $b \in B' - A'$, if the closure of A' is in the interior of B' , and if for some natural numbers $m \geq 2$ and $n \geq 1$, we have that $\pi_j(X' - A', B' - A') = 0$ for all $j < m$ and $\pi_j(B', B' - A') = 0$ for all $j < n$, then for all $j < n + m - 2$,

$$\pi_j(X' - A', B' - A') \rightarrow \pi_j(X', B')$$

is an isomorphism. We can now apply this to our case: since we know that $\Sigma^i Y$ is $i - 1$ connected, and we know that taking smash products can only increase connectivity, we can deduce that

$$\pi_j((X - A) \wedge \Sigma^i Y, (B - A) \wedge \Sigma^i Y) = 0 = \pi_j(B \wedge \Sigma^i Y, (B - A) \wedge \Sigma^i Y)$$

for all $j < i - 1$. Writing $A' := A \wedge \Sigma^i Y$, $B' := B \wedge \Sigma^i Y$ and $X' := X \wedge \Sigma^i Y$, we see that the conditions of the homotopy excision theorem are satisfied for $n = m = i - 1$, which allows us to conclude that

$$\pi_{k+i}((X - A) \wedge \Sigma^i Y, (B - A) \wedge \Sigma^i Y) \rightarrow \pi_{k+i}(X \wedge \Sigma^i Y, B \wedge \Sigma^i Y)$$

is an isomorphism if $k + i < 2i - 4$. This means if $i > k + 4$, excision holds, which is what we wanted.

Now we can extend this result to all spectra: let $Y := (Y_i, \gamma_i)$ be an arbitrary spectrum. We can define $T^n Y$ to be the spectrum which is identical to Y up to the index n , above which we have $(T^n Y)_i := \Sigma^{i-n} Y_n$ with the structure maps being the identity. The homotopy groups of $X \wedge T^n Y$ are the same as the homotopy groups of $X \wedge \Sigma^\infty Y_n$ up to a shift of indices. This means $H_*^{T^n Y}$ is also a homology theory. The rest of the argument can be explained with a single diagram:

$$\begin{array}{ccccccc}
& \vdots & & \vdots & & \vdots & \\
& \uparrow & & \uparrow & & \uparrow & \\
\pi_{k+2}(X \wedge \Sigma^2 Y_0) & \longrightarrow & \pi_{k+2}(X \wedge \Sigma Y_1) & \longrightarrow & \pi_{k+2}(X \wedge Y_2) & \longrightarrow & \dots \\
& \uparrow & & \uparrow & & \uparrow & \\
\pi_{k+1}(X \wedge \Sigma Y_0) & \longrightarrow & \pi_{k+1}(X \wedge Y_1) & \longrightarrow & \pi_{k+1}(X \wedge Y_1) & \longrightarrow & \dots \\
& \uparrow & & \uparrow & & \uparrow & \\
\pi_k(X \wedge Y_0) & \longrightarrow & \pi_k(X \wedge Y_0) & \longrightarrow & \pi_k(X \wedge Y_0) & \longrightarrow & \dots
\end{array}$$

Suppose we take the colimit of this diagram. If we first take the colimit of each row, it will give us $\pi_{k+i}(X \wedge Y_i)$ for the i th row. Taking the colimit of the result is by definition $\pi_k(X \wedge Y)$.

On the other hand, if we take the colimit of the columns first, the i th column gives us $\pi_k(X \wedge T^i Y)$, which means $\pi_k(X \wedge Y)$ is the colimit of the $\pi_k(X \wedge T^i Y)$ s. This means we can write H^Y as the colimit of the chain $H^{T^i Y}$ of homology theories. As the excision maps $H^{T^i Y}(X - A, B - A) \rightarrow H^{T^i Y}(X, B)$ are isomorphisms, the excision map on the colimit must also be an isomorphism, which means H^Y is a homology theory, as desired. This completes the proof.

4.3 Spectra Induce Cohomology Theories

We will now shift our focus towards cohomology. In this section, we will explain how spectra can be used to represent cohomology theories, and in the process define a natural dual to configuration homology: a cohomology theory we will call configuration cohomology. We will explore configuration cohomology in detail in the next section.

We start with a definition:

Definition 4.12 (Stable Maps). Let X be a based topological space, and let Y be a spectrum. The k -th stable class of maps $[X, Y]_k$ is the colimit of the sequence

$$[\Sigma^k X, Y_0] \rightarrow [\Sigma^{k+1} X, \Sigma Y_0] \rightarrow [\Sigma^{k+1} X, Y_1] \rightarrow \dots$$

in the category of abelian groups, where we ignore the first $-2k+3$ terms (as they are either undefined or do not have a natural abelian group structure). $\triangle$

Note that the homotopy groups of a spectrum Y are a special case of this: namely, $\pi_k(Y) = [S^0, Y]_k$. Our goal in this section will be to show that the contravariant functors H_Y^k which send a based space X to the abelian group $[X, Y]_{-k}$ form a reduced cohomology theory on CW-complexes. To do this, we will use an alternative to the Eilenberg-Steenrod axioms for a reduced cohomology theory. A reduced cohomology theory satisfying these alternate axioms is called a Whitehead Generalized Cohomology Theory. [7, p. 251-252] In essence,

the axioms of homotopy invariance, additivity, and the long exact sequence remain the same, but instead of excision, we define relative cohomology groups to be $\tilde{H}^k(X, A) := \tilde{H}^k(C(X, A))$, where $C(X, A) := CA \cup X \times \{1\}$. The long exact sequence in cohomology is then the only interesting axiom to verify. For all pairs of based spaces (X, A) with $b \in A \subset X$, we need to construct natural homomorphisms (called coboundary maps) $\partial^* : H_Y^*(A) \rightarrow H_Y^{*+1}(C(X, A))$ which induce the long exact sequence of (reduced) cohomology for (X, A) .

These coboundary maps will be induced by the map $C(X, A) \rightarrow \Sigma A$ which collapses $X \times \{1\}$. Specifically, this map gives us the commutative diagram

$$\begin{array}{ccccccc} [\Sigma^{-k} A, Y_0] & \longrightarrow & [\Sigma^{-k+1} A, Y_1] & \longrightarrow & [\Sigma^{-k+2} A, Y_2] & \longrightarrow & \dots \\ \downarrow & & \downarrow & & \downarrow & & \\ [\Sigma^{-k-1} C(X, A), Y_0] & \longrightarrow & [\Sigma^{-k} C(X, A), Y_1] & \longrightarrow & [\Sigma^{-k+1} C(X, A), Y_2] & \longrightarrow & \dots \end{array}$$

which induces our map $\partial_Y^k : H_Y^k(A) \rightarrow H_Y^{k+1}(C(X, A))$. Now we just need to show this gives a long exact sequence

$$\dots \rightarrow H_Y^{k-1}(A) \rightarrow H_Y^k(C(X, A)) \rightarrow H_Y^k(X) \rightarrow H_Y^k(A) \rightarrow H_Y^{k+1}(C(X, A)) \rightarrow \dots$$

This sequence arises from taking the colimit of each column in the diagram

$$\begin{array}{ccccccc} & & \vdots & & \vdots & & \vdots \\ & & \uparrow & & \uparrow & & \uparrow \\ \dots & \longrightarrow & [\Sigma^{-k+2} C(X, A), Y_2] & \longrightarrow & [\Sigma^{-k+2} X, Y_2] & \longrightarrow & [\Sigma^{-k+2} A, Y_2] \longrightarrow \dots \\ & & \uparrow & & \uparrow & & \uparrow \\ \dots & \longrightarrow & [\Sigma^{-k+1} C(X, A), Y_1] & \longrightarrow & [\Sigma^{-k+1} X, Y_1] & \longrightarrow & [\Sigma^{-k+1} A, Y_1] \longrightarrow \dots \\ & & \uparrow & & \uparrow & & \uparrow \\ \dots & \longrightarrow & [\Sigma^{-k} C(X, A), Y_0] & \longrightarrow & [\Sigma^{-k} X, Y_0] & \longrightarrow & [\Sigma^{-k} A, Y_0] \longrightarrow \dots \end{array}$$

and since exactness is preserved by colimits, all we need to show is that each row is exact as a sequence of pointed sets. Up to a shift of indices, after ignoring undefined terms (i.e. negative suspensions) each row is of the form

$$\dots \rightarrow [\Sigma X, Z] \rightarrow [\Sigma A, Z] \rightarrow [C(X, A), Z] \rightarrow [X, Z] \rightarrow [A, Z]$$

for some space Z . As replacing (X, A) by $(\Sigma X, \Sigma A)$ has the effect of shifting this row to the left, to show this whole row is exact, it is sufficient to show that the sequence

$$[\Sigma X, Z] \rightarrow [\Sigma A, Z] \rightarrow [C(X, A), Z] \rightarrow [X, Z] \rightarrow [A, Z]$$

is exact. Let

$$A \xrightarrow{f_1} X \xrightarrow{f_2} C(X, A) \xrightarrow{f_3} \Sigma A \xrightarrow{\Sigma f_1} \Sigma X$$

be the maps that induce our sequence. We can then show exactness pairwise: for the first pair, $f_2 \circ f_1$ is the constant map, so $f_1^* \circ f_2^* : [C(X, A), Z] \rightarrow [A, Z]$ must also be constant. Conversely, suppose $g : X \rightarrow Z$ is a map such that $g \circ f_1$ is null-homotopic. By definition that means g extends to a map on $C(X, A)$, so $[g]$ is in the image of f_2^* , which shows exactness of the first pair.

For the second pair, we again have that $f_3 \circ f_2$ is constant, so we only need to show that the kernel of f_2^* is in the image of f_3^* . So, let $g : C(X, A) \rightarrow Z$ be a map such that becomes null-homotopic when restricted to $X \times \{1\}$. That means g can be extended to a map $g' : J \rightarrow Z$, where $J := (A \times [0, 2] \cup X \times [1, 2]) / (\{b\} \times [0, 2] \cup A \times \{0\} \cup X \times \{2\})$. We can then define a homotopy $H : C(X, A) \rightarrow Z$ of g to be $H([(x, s)], t) := g'(x, s(1+t))$. The end result of this homotopy is a map that sends $X \times \{1\}$ to the basepoint, which means it is in the image of f_3^* as desired.

Finally, we will show the exactness of the third pair. We start by showing that $\Sigma f_1 \circ f_3$ is null-homotopic by explicitly constructing a homotopy $H : C(X, A) \times I \rightarrow \Sigma X$ defined as $H([(x, s)], t) := [(x, \max(s-t, 0))]$. We can manually confirm this is a valid null-homotopy, as desired. This means all we have left to show is that the kernel of f_3^* is in the image of Σf_1^* .

So, suppose $g : \Sigma A \rightarrow Z$ is a map such that $g \circ f_3$ is null-homotopic. Unwinding definitions, this means we have a map $K : A \times I^2 \cup X \times \{1\} \times I \rightarrow Z$ that is g on the face $A \times I \times \{0\} \cup X \times \{1\} \times \{0\}$, and maps to the basepoint on the faces $A \times \{0\} \times I$ and $A \times I \times \{1\} \cup X \times \{1\} \times \{1\}$. We can from this construct an explicit homotopy $H : C(X, A) \times I \rightarrow Z$ of g defined as

$$H([(x, s)], t) := \begin{cases} K(x, s, 1-2t) & \text{if } 2t \leq 1-s \\ K(x, \min(2t+s-1, 1), s) & \text{otherwise.} \end{cases}$$

which is continuous by the pasting lemma. The end result of this homotopy will be a map that, by the definition of K , will extend to a map on ΣX as desired. We have now proven that H_Y^* is a cohomology theory.

Previously, we have shown that the homology theory H_*^Y has a nice interpretation when Y is an Ω -spectrum. An analogous statement holds for H_Y^* :

Proposition 1. *Let Y be an Ω -spectrum. Then on CW-complexes, for all non-negative k , H_Y^k is naturally isomorphic to the functor $X \mapsto [X, Y_k]$, and for all negative k , it is naturally isomorphic to $X \mapsto [X, \Omega^{-k} Y_0]$. If on top of this Y is -1 -connected, then we also have that $H_Y^k(S^0)$ is trivial for $k > 0$.*

Proof. For the first part, it is enough to show that each map $[\Sigma^{-k+i} X, Y_i] \rightarrow [\Sigma^{-k+i+1} X, Y_{i+1}]$ in the defining colimit of $H_Y^k(X)$ is an isomorphism. To see why, note that for positive k this is sufficient because it implies the map $[X, Y_k] \rightarrow H_Y^k(X)$ is an isomorphism, and for negative k it implies that $[X, \Omega^{-k} Y_0] \cong [\Sigma^{-k} X, Y_0] \rightarrow H_Y^k(X)$ is an isomorphism, as desired.

Now, the map $[\Sigma^{-k+i}X, Y_i] \rightarrow [\Sigma^{-k+i+1}X, Y_{i+1}] \cong [\Sigma^{-k+i}X, \Omega Y_{i+1}]$ is induced by the structure map $Y_i \rightarrow \Omega Y_{i+1}$ which is a weak equivalence by definition. Any weak equivalence $Z_1 \rightarrow Z_2$ induces a bijection on the mapping classes $[X, Z_1] \rightarrow [X, Z_2]$ if X is a CW-complex, so we are done.

Finally, if Y is -1 connected, then by definition the mapping classes $[S^0, X_k]$ will be trivial for $k > 0$. This completes the proof. $\square$

We can immediately derive two more corollaries from this:

Corollary 4.13. *If Y is an Eilenberg-MacLane spectrum, then $H^k(X) := [X, Y_k]$ defines an ordinary cohomology theory.*

Corollary 4.14. *For any discrete abelian group G , $H^k(X) := [X, G_0^{(S^k)}]$ defines an ordinary cohomology theory with coefficients in G .*

We end this section by turning this last corollary into a definition:

Definition 4.15 (Configuration Cohomology). Let X be a space, and let G be a discrete abelian group. We define the *configuration cohomology* $H_{\text{conf}}^k(X; G)$ of X with coefficients in G to be $[X, G^{(S^k)}] \cong H_{G(\Sigma^\infty_{S^0})}^k(X)$ for each k . $\triangle$

5 Configuration Cohomology

In this section, we will explore how configuration cohomology, as defined at the end of the last section, can be used to represent cohomology operations.

We start with a definition:

Definition 5.1 (Cohomology Operation). For $0 \leq i \leq k$, let n_i be integers and let G_i be abelian groups. Let n' and G' also be an integer and an abelian group respectively. Then a *cohomology operation* consists of, for all spaces X , a function (not necessarily a homomorphism)

$$\prod_{i=1}^k H^{n_i}(X; G_i) \rightarrow H^{n'}(X; G')$$

that is natural in X . $\triangle$

Before proceeding further, we will provide a brief overview of some of the most common examples of cohomology operations:

- For any abelian group G and natural number n , addition $+: H^n(X; G) \times H^n(X; G) \rightarrow H^n(X; G)$ is a cohomology operation.
- For any group homomorphism $f: G \rightarrow H$ and natural number n , the induced map $H^n(X; G) \rightarrow H^n(X; H)$ is a cohomology operation. If $n = 0$, then we may assume f to be any function, not just a homomorphism.
- Given a ring R , the cup product $\smile: H^n(X; R) \times H^m(X; R) \rightarrow H^{n+m}(X; R)$ is a cohomology operation.

- Given a short exact sequence $N \hookrightarrow G \twoheadrightarrow G/N$ of abelian groups, the long exact sequence in cohomology associated to the induced short exact sequence of cochain complexes $C_{\text{sing}}^*(X; N) \hookrightarrow C_{\text{sing}}^*(X; G) \twoheadrightarrow C_{\text{sing}}^*(X; G/N)$ gives homomorphisms $H^n(X; G/N) \rightarrow H^{n+1}(X; N)$ for each n . These cohomology operations are called Bockstein maps.
- There exists a series of cohomology operations $\text{Sq}^i : H^n(X; \mathbb{Z}_2) \rightarrow H^{n+i}(X; \mathbb{Z}_2)$ called the Steenrod Squares [4, p. 487-518], where both n and i are natural numbers. While we will not explicitly define them for brevity, we can axiomatically characterize them using a few short rules:
 1. Each Sq^i is a cohomology operation.
 2. The map Sq^0 is the identity.
 3. If $|x| = n$, then $\text{Sq}^n(x) = x \smile x$.
 4. If $\text{Sq} := \text{Sq}^0 + \text{Sq}^1 + \text{Sq}^2 + \dots$ is treated as an operation on the cohomology ring $H^*(X; \mathbb{Z}_2)$, then $\text{Sq}(x \smile y) = \text{Sq}(x) \smile \text{Sq}(y)$ for all $x, y \in H^*(X; \mathbb{Z}_2)$.

Our goal in this section will be to geometrically represent as many of these cohomology operations in terms of configuration cohomology as we can. To achieve this, we will use the following construction: We know that the functor $X \mapsto H^n(X; G)$ is naturally isomorphic to the functor $X \mapsto [X, G_0^{(S^n)}]$, where both functors have **hTop** as the domain (the homotopy category of topological spaces) and **Sets** as the codomain. A cohomology operation is then a natural transformation of functors

$$f_X : [X, \prod_{i=1}^m G_{i,0}^{(S^{n_i})}] \rightarrow [X, G_0'^{(S^{n'})}].$$

The Yoneda lemma then tells us that this cohomology operation is induced by a map in $[\prod_{i=1}^m G_{i,0}^{(S^{n_i})}, G_0'^{(S^{n'})}]$. More specifically, it tells us that if $\alpha \in [X, \prod_{i=1}^m G_{i,0}^{(S^{n_i})}]$, then

$$f_X(\alpha) = f_{\prod_{i=1}^m G_{i,0}^{(S^{n_i})}}(\text{id}_{\prod_{i=1}^m G_{i,0}^{(S^{n_i})}}) \circ \alpha.$$

In summary, the Yoneda lemma gives us a one-to-one correspondence between cohomology operations $\prod_{i=1}^m H^{n_i}(X; G_i) \rightarrow H^{n'}(X; G')$ and homotopy-classes of maps $\prod_{i=1}^m G_{i,0}^{(S^{n_i})} \rightarrow G_0'^{(S^{n'})}$. If a map $\prod_{i=1}^m G_{i,0}^{(S^{n_i})} \rightarrow G_0'^{(S^{n'})}$ relates to the cohomology operation $\prod_{i=1}^m H^{n_i}(X; G_i) \rightarrow H^{n'}(X; G')$ under this correspondence, then we say it represents the cohomology operation.

Our goal for the rest of this section will be to find explicit representations for the cohomology operations we described above.

5.1 Representing Basic Cohomology Operations

In this section we will represent the cohomology operation of addition, and both classify and represent all cohomology operations of the form $H^n(X; G) \rightarrow H^n(X; H)$. These more basic cohomology operations are especially important for our purposes, as they are useful when constructing more complicated cohomology operations.

It is no surprise that addition, the cohomology operation $H^n(X; G) \times H^n(X; G) \rightarrow H^n(X; G)$, is represented by addition, the map $G_0^{(S^n)} \times G_0^{(S^n)} \rightarrow G_0^{(S^n)}$. Note that this is not immediate: in the definition of configuration cohomology, the group structure of $H_{\text{conf}}^n(X)$ arises from the weak equivalence $G^{(S^n)} \rightarrow \Omega G^{(S^{n+1})}$, not from the group structure of $G^{(S^n)}$. We can show that these two group structures are equivalent by showing that the diagram

$$\begin{array}{ccc} G_0^{(S^n)} \times G_0^{(S^n)} & \longrightarrow & G_0^{(S^n)} \\ \downarrow & & \downarrow \\ \Omega G_0^{(S^{n+1})} \times \Omega G_0^{(S^{n+1})} & \longrightarrow & \Omega G_0^{(S^{n+1})} \end{array}$$

commutes up to homotopy, where the top map is addition and the bottom map is composition of loops. The bottom map is homotopic to the pointwise addition map, and applying that homotopy makes the diagram commute, so $+: G_0^{(S^n)} \times G_0^{(S^n)} \rightarrow G_0^{(S^n)}$ does indeed represent addition.

We will now classify and represent all cohomology operations of the form $H^n(X; G) \rightarrow H^n(X; H)$. First, we will examine the special case where $n = 0$: Cohomology operations $H^0(X; G) \rightarrow H^0(X; H)$ correspond one-to-one with homotopy-classes of maps $[G_0^{(S^0)}, H_0^{(S^0)}]$. Since $G_0^{(S^0)} \cong G$ and $H_0^{(S^0)} \cong H$ are both discrete spaces, $[G_0^{(S^0)}, H_0^{(S^0)}]$ is equal to the set of all functions $G \rightarrow H$, which means cohomology operations $H^0(X; G) \rightarrow H^0(X; H)$ are both classified and represented by functions $G \rightarrow H$.

Now assume that $n > 0$. Then cohomology operations $H^n(X; G) \rightarrow H^n(X; H)$ correspond to elements of $[G_0^{(S^n)}, H_0^{(S^n)}]$, which are elements in $H^n(G_0^{(S^n)}; H)$. The universal coefficient theorem, alongside the fact that $G_0^{(S^n)}$ is $n-1$ -connected, tells us that $H^n(G_0^{(S^n)}; H) \cong \text{Hom}(H_n(G_0^{(S^n)}; \mathbb{Z}), H)$. The Hurewicz theorem then tells us that $H_n(G_0^{(S^n)}; \mathbb{Z}) \cong \pi_n(G_0^{(S^n)}) \cong G$, which means $[G_0^{(S^n)}, H_0^{(S^n)}] \cong \text{Hom}(G, H)$. We can assign an explicit representation $\varphi' : G_0^{(S^n)} \rightarrow H_0^{(S^n)}$ to an element $\varphi \in \text{Hom}(G, H)$ by applying φ to the coefficients of a formal sum in $G_0^{(S^n)}$. To see that this assignment is compatible with the isomorphism $[G_0^{(S^n)}, H_0^{(S^n)}] \cong \text{Hom}(G, H)$ we just constructed, it is sufficient to manually note that the following diagram

$$\begin{array}{ccc} \pi_n(G_0^{(S^n)}) & \xrightarrow{\varphi'_*} & \pi_n(H_0^{(S^n)}) \\ \uparrow & & \uparrow \\ G & \xrightarrow{\varphi} & H \end{array}$$

commutes, where the vertical maps come from the structure maps $G \rightarrow \Omega^n G_0^{(S^n)}$ and $H \rightarrow \Omega^n H_0^{(S^n)}$ respectively.

We are now ready to represent more complicated cohomology operations.

5.2 The Cup Product

In this section, we will represent the cup product $H^n(X; R) \times H^m(X; R) \rightarrow H^{n+m}(X; R)$ in terms of configuration cohomology. We will first give an explicit formula: For any ring R and a pair of spaces X and Y , we can define a map $R_0^{(X)} \times R_0^{(Y)} \rightarrow R_0^{(X \times Y)}$ sending each $(\sum_{i=1}^n r_i x_i, \sum_{j=1}^m r'_j y_j)$ to $\sum_{i=1}^n \sum_{j=1}^m r_i r'_j (x_i, y_j)$ (note that this map is continuous by the fundamental property of quotients). We can then represent the cup product in configuration cohomology as the composition of maps

$$\smile_{\text{conf}}: R_0^{(S^n)} \times R_0^{(S^m)} \rightarrow R_0^{(S^n \times S^m)} \rightarrow R_0^{(S^n \wedge S^m)} \cong R_0^{(S^{n+m})}$$

where $R_0^{(S^n \times S^m)} \rightarrow R_0^{(S^n \wedge S^m)}$ is induced by the quotient map $S^n \times S^m \rightarrow S^n \wedge S^m$. Before we prove that $\smile_{\text{conf}}$ represents the cup product, we will first confirm it has properties we would reasonably expect from a cup product:

1. The operation $\smile_{\text{conf}}$ is associative. We can check this manually from the definition (note that it is actually strictly associative, not just up to homotopy).
2. If X is connected and $r \in H_{\text{conf}}^0(X; R)$, then for any $\alpha \in H_{\text{conf}}^n(X; R)$, the map $\alpha \mapsto r \smile_{\text{conf}} \alpha$ acts as multiplication by r . We can check this manually as well: Given $r \in R \cong R_0^{(S^0)}$ and $s = \sum_{i=1}^k r'_i x_i \in R_0^{(S^n)}$, we have that $r \smile_{\text{conf}} s = rs$.
3. The operation $\smile_{\text{conf}}$ is graded commutative. This is because the smash product map on spheres is graded commutative [4, p. 388].

While this does not prove that $\smile_{\text{conf}}$ represents the cup product, it still gives us a strong geometric intuition as to why the cup product satisfies these properties.

To prove that $\smile_{\text{conf}}$ represents the usual cup product, we will instead show that the formula for the cross product $\times_{\text{conf}}: H_{\text{conf}}^n(X; R) \times H_{\text{conf}}^m(Y; R) \rightarrow H_{\text{conf}}^{n+m}(X \times Y; R)$ induced by $\smile_{\text{conf}}$ aligns with the cellular description of the cross product (which can be found in [4, p. 278-280]). Note that this is both necessary and sufficient: any cup product always induces a cross product and vice-versa. We will first reduce the problem to spheres: Let X and Y be CW-complexes, and let n be a natural number. Then the following diagram

$$\begin{array}{ccc}
\bigoplus_{i+j=n} H^i(X; R) \times H^j(Y; R) & \longrightarrow & H^n(X \times Y; R) \\
\downarrow & & \downarrow \\
\bigoplus_{i+j=n} H^i(X_i; R) \times H^j(Y_j; R) & \longrightarrow & \bigoplus_{i+j=n} H^n(X_i \times Y_j; R) \\
\uparrow & & \uparrow \\
\bigoplus_{i+j=n} H^i(X_i/X_{i-1}; R) \times H^j(Y_j/Y_{j-1}; R) & \longrightarrow & \bigoplus_{i+j=n} H^n(X_i/X_{i-1} \times Y_j/Y_{j-1}; R)
\end{array}$$

commutes, where the horizontal maps are induced by the cross product in configuration cohomology. Note that due to the injectivity of the top vertical maps and the surjectivity of the bottom vertical maps (these maps are injective and surjective respectively by the definition of cellular cohomology), the cross product map on the bottom induces the cross product map on the top. To put it another way: if we know how the cross product behaves on wedge-sums of spheres (or, in the zero-dimensional case, disjoint unions of points), then we know how it behaves everywhere.

We can reduce even further: since $\times_{\text{conf}}$ respects both disjoint unions and wedge-sums, it is enough to check that $\times_{\text{conf}} : H^n(S^n; R) \times H^m(S^m; R) \rightarrow H^{n+m}(S^n \times S^m; R)$ aligns with the cellular cross product whenever $n, m > 0$ (as we have already treated the case where one of them is zero separately). We can compose $\times_{\text{conf}}$ with the isomorphism $H^{n+m}(S^n \times S^m) \rightarrow H^{n+m}(S^n \wedge S^m) \cong H^{n+m}(S^{n+m})$ to get a map $H^n(S^n; R) \times H^m(S^m; R) \rightarrow H^{n+m}(S^{n+m}; R)$. According to the cellular description of the cross product, this map should be the multiplication map $R \times R \rightarrow R$ when $H^n(S^n; R), H^m(S^m; R)$ and $H^{n+m}(S^{n+m}; R)$ are all identified with R . To confirm this holds for $\times_{\text{conf}}$ we define, for all natural numbers i and $r \in R$, a map $\alpha_r^i : S^i \rightarrow R_0^{(S^i)}, x \mapsto r(x - b)$ where b is the basepoint of S^i . As we have seen in Section 5.1, α_r^i is a representative of $r \in H^i(S^i; R)$. We can manually confirm that the map $H^n(S^n; R) \times H^m(S^m; R) \rightarrow H^{n+m}(S^{n+m}; R)$ sends $(\alpha_{r_1}^n, \alpha_{r_2}^m)$ to $\alpha_{r_1 r_2}^{n+m}$, which means $\times_{\text{conf}}$ coincides with the cellular cross product on spheres, as desired.

This proves that $\smile_{\text{conf}}$ represents the cup product.

5.3 A Representation of Sq^1

In the introduction to this section, we listed the Bockstein maps and the Steenrod Squares as important examples of cohomology operations. Constructing a representation of each of these is beyond the scope of this thesis. However, we will construct a representation of the Bockstein map for the short exact sequence

$$0 \rightarrow \mathbb{Z}_2 \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2 \rightarrow 0,$$

which happens to be $\text{Sq}^1 : H^n(X; \mathbb{Z}_2) \rightarrow H^{n+1}(X; \mathbb{Z}_2)$.

We have previously defined the Bockstein maps in terms of singular cohomology. We will now present an alternative definition of the Bockstein maps in terms of Eilenberg-MacLane spectra, which will be more useful for our purposes: Let

$$0 \rightarrow N \rightarrow G \rightarrow G/N \rightarrow 0$$

be a short exact sequence of abelian groups, and let

$$K(N, n) \rightarrow K(G, n) \rightarrow K(G/N, n)$$

be corresponding maps between Eilenberg-MacLane spaces (i.e. these maps represent the cohomology operations induced by the maps $N \rightarrow G \rightarrow G/N$). We may assume without loss of generality that $K(G, n) \rightarrow K(G/N, n)$ is a fibration. We will also assume that these spaces form Eilenberg-MacLane spectra, and that the diagram

$$\begin{array}{ccccc} \Sigma K(N, n) & \longrightarrow & \Sigma K(G, n) & \longrightarrow & \Sigma K(G/N, n) \\ \downarrow & & \downarrow & & \downarrow \\ K(N, n+1) & \longrightarrow & K(G, n+1) & \longrightarrow & K(G/N, n+1) \end{array}$$

commutes for all n up to homotopy. Since $K(G, n) \rightarrow K(G/N, n)$ is a fibration, we can find a map $f : K(G/N, n) \times I \rightarrow K(G, n+1)$ that makes the diagram

$$\begin{array}{ccc} K(G/N, n) \times \{0\} & \longrightarrow & K(G, n+1) \\ \downarrow & \nearrow f & \downarrow \\ K(G/N, n) \times I & \longrightarrow & K(G/N, n+1) \end{array}$$

commute, where the top map sends every point to the basepoint, and the bottom map is the composition $K(G/N, n) \times I \rightarrow \Sigma K(G/N, n) \rightarrow K(G/N, n+1)$. Now, by definition $f(K(G/N, n) \times \{1\}) \subset K(N, n+1)$, which gives us a map $K(G/N, n) \rightarrow K(N, n+1)$. This map represents the Bockstein homomorphism of the sequence $N \rightarrow G \rightarrow G/N$.

Before we proceed, we will quickly show that the two definitions of the Bockstein map we presented (which we will call the “singular” and “spectral” definitions for this proof) align: First, note that we can translate the singular definition into a cellular one by just switching to cellular cochain complexes. By the Yoneda lemma, it is sufficient to show that the cellular definition is correct on the fundamental class $K(G/N, n) \rightarrow K(G/N, n)$ (we may assume each Eilenberg-MacLane space we use to be a CW complex). Now, the cellular description of the Bockstein map arises from the long exact sequence of cellular cochains, which uses the coboundary map $H^n(K(G, n)_n, K(G, n)_{n-1}) \rightarrow H^{n+1}(K(G, n)_{n+1}, K(G, n)_n)$, which itself arises from the structure map $\Sigma K(G, n) \rightarrow K(G, n+1)$. In contrast, the spectral definition of the Bockstein map first applies the structure map $\Sigma K(G/N, n) \rightarrow K(G/N, n+1)$ and then lifts up to $K(G, n+1)$. We have assumed these two approaches commute up to homotopy, so it is a matter of unwinding definitions to see that the cellular definition and the spectral definition of the Bockstein map agree on the fundamental class.

We will now apply the spectral definition to the defining Eilenberg-MacLane spectrum of configuration cohomology: Let

$$N_0^{(S^n)} \rightarrow G_0^{(S^n)} \rightarrow (G/N)_0^{(S^n)}$$

be the maps induced by the short exact sequence $N \rightarrow G \rightarrow G/N$. We can manually check that the diagram

$$\begin{array}{ccccc} \Sigma N_0^{(S^n)} & \longrightarrow & \Sigma G_0^{(S^n)} & \longrightarrow & \Sigma (G/N)_0^{(S^n)} \\ \downarrow & & \downarrow & & \downarrow \\ N_0^{(S^{n+1})} & \longrightarrow & G_0^{(S^{n+1})} & \longrightarrow & (G/N)_0^{(S^{n+1})} \end{array}$$

commutes (strictly in this case, not just up to homotopy). However, in order to be able to apply the definition of the Bockstein homomorphism to these formal sum spaces, for the remainder of this section, we will need to assume the following:

Conjecture 5.2. *Let $G \rightarrow G/N$ be a surjective homomorphism between abelian groups. Then for all n , the map*

$$G_0^{(S^n)} \rightarrow (G/N)_0^{(S^n)}$$

is a fibration with fiber $N_0^{(S^n)}$.

Assuming this conjecture, we will be able to construct Sq^1 by explicitly constructing the diagonal map in the diagram

$$\begin{array}{ccc} (G/N)_0^{(S^n)} \times \{0\} & \xrightarrow{0} & G_0^{(S^{n+1})} \\ \downarrow & \nearrow & \downarrow \\ (G/N)_0^{(S^n)} \times I & \longrightarrow & (G/N)_0^{(S^{n+1})} \end{array}$$

in the case where $G = \mathbb{Z}_4$ and $N = \mathbb{Z}_2$.

Before we construct $\mathbb{Z}_{2,0}^{(S^n)} \times I \rightarrow \mathbb{Z}_{4,0}^{(S^{n+1})}$, we need to do some groundwork: Let $L_n = \{(x, y) \in S^n \times S^n \mid x \neq -y\}$, where S^n is treated as a subspace of $\mathbb{R}^{n+1}$. We can define the map $c : L_n \rightarrow S^n$ via the rule $c(x, y) = \frac{x+y}{|x+y|}$. This map has the following two properties that are important for our purposes:

1. The map c is symmetric: for any $(x, y) \in L_n$, $c(x, y) = c(y, x)$
2. For any $x \in S^n$, $c(x, x) = x$.

We will also define the map $d : S^n \times S^n \rightarrow I$ via the rule

$$d(x, y) := \begin{cases} |x - y| & \text{if } |x - y| \leq \frac{1}{2} \\ \frac{1}{2} & \text{otherwise.} \end{cases}$$

We can use these two maps together to define a map

$$J : S^n \times S^n \times I \rightarrow \mathbb{Z}_{4,0}^{(S^n \times I)}$$

such that for all $(x, y, t) \in S^n \times S^n \times I$,

$$J(x, y, t) := \begin{cases} 2(c(x, y), td(x, y)) + 2(c(x, y), t(1 - d(x, y))) & \text{if } x \neq -y, \\ 0 & \text{otherwise.} \end{cases}$$

We can use the pasting lemma to show that J is continuous, as it is continuous if we restrict the domain to $|x - y| \leq \frac{1}{2}$, and the constant 0 if we restrict it to $|x - y| \geq \frac{1}{2}$. Note the following:

1. Due to the fact that c and d are both symmetric, we have that $J(x, y, t) = J(y, x, t)$ for all $x, y \in S^n$ and $t \in I$.
2. Since $c(x, x) = x$ for all $x \in S^n$, we have that $J(x, x, t) = 2(x, 0) + 2(x, t)$.

We will now define the map $f' : \mathbb{Z}_{2,0}^{(S^n)} \times I \rightarrow \mathbb{Z}_{4,0}^{(S^n \times I)}$ via the rule

$$f'(\sum_{i=1}^m x_i, t) := \sum_{i=1}^m ((x_i, 0) - (x_i, t)) + \sum_{1 \leq i < j \leq m} J(x_i, x_j, t)$$

for all $\sum_{i=1}^m x_i \in \mathbb{Z}_{2,0}^{(S^n)}$ and $t \in I$ (see Figure 8 for a visual representation of f'). Now, if f' is well defined, then it is also continuous by the fundamental property of quotients. We will show that it is well defined using arguments similar to the ones we used in Section 2.3: Let $\sum_{i=1}^{m_1} x_i = \sum_{j=1}^{m_2} y_j \in \mathbb{Z}_{2,0}^{(S^n)}$, and suppose that m_2 is minimal (i.e. m_2 is the minimal amount of summands required to express $\sum_{i=1}^{m_1} x_i$). Then we will show that $f'(\sum_{i=1}^{m_1} x_i) = f'(\sum_{j=1}^{m_2} y_j)$ by induction on m_1 : in the base case, $m_1 = m_2$, which means the sums are identical up to a permutation of summands. Since J is symmetric, the definition of f' does not depend on the order of the summands, so the base case holds. Now suppose $m_1 > m_2$. That means that at least two summands in $\sum_{i=1}^{m_1} x_i$ are identical. Without loss of generality, we may assume $x_{m_1} = x_{m_1-1}$, in which case we have

$$f'(\sum_{i=1}^{m_1} x_i, t) - f'(\sum_{i=1}^{m_1-2} x_i, t) = 2(x_{m_1}, 0) + 2(x_{m_1}, t) + J(x_{m_1}, x_{m_1}, t) + \sum_{i=1}^{m_1-2} 2J(x_i, x_{m_1}, t)$$

Now, each $2J(x_i, x_{m_1}, t) = 0$ as J only maps to 2-torsion elements, and we also have that

$$2(x_{m_1}, 0) + 2(x_{m_1}, t) + J(x_{m_1}, x_{m_1}, t) = 0,$$

so $f'(\sum_{i=1}^{m_1} x_i, t) = f'(\sum_{i=1}^{m_1-2} x_i, t)$ as desired. This proves that f' is well defined.

We can compose f' with the map $\mathbb{Z}_{4,0}^{(S^n \times I)} \rightarrow \mathbb{Z}_{4,0}^{(\Sigma S^n)} \cong \mathbb{Z}_{4,0}^{(S^{n+1})}$ to get a map $f : \mathbb{Z}_{2,0}^{(S^n)} \times I \rightarrow \mathbb{Z}_{4,0}^{(S^{n+1})}$, and we can manually check that f makes the diagram

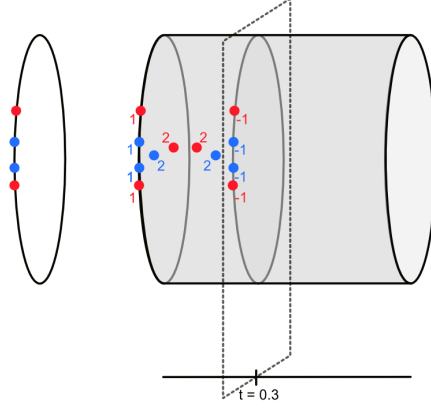


Figure 8: An illustration of f' for $n = 1$ with $t = 0.3$. The red formal sum maps to the red formal sum, and the blue formal sum maps to the blue formal sum. We can see that the closer the initial pair of summands are, the smaller “triangles” they form, which cancel each other out if the summands collide. If the initial summands get far away from each other, then the two central summands in the sum they map to approach each other in the middle until they cancel each other out.

$$\begin{array}{ccc}
 \mathbb{Z}_{2,0}^{(S^n)} \times \{0\} & \xrightarrow{0} & \mathbb{Z}_{4,0}^{(S^{n+1})} \\
 \downarrow & \nearrow f & \downarrow \\
 \mathbb{Z}_{2,0}^{(S^n)} \times I & \longrightarrow & \mathbb{Z}_{2,0}^{(S^{n+1})}
 \end{array}$$

commute. To attain from this a representation $\text{Sq}^1 : \mathbb{Z}_{2,0}^{(S^n)} \rightarrow \mathbb{Z}_{2,0}^{(S^{n+1})}$ of the Bockstein map, we first need to examine the behaviour of f at $t = 1$: Treating S^{n+1} as $\Sigma S^n = S^n \times I / \sim$, we have that

$$f\left(\sum_{i=1}^m x_i, 1\right) = \sum_{1 \leq i < j \leq m} [J(x_i, x_j, 1)].$$

To get a formula for Sq^1 , we need to lift f to the fiber $\mathbb{Z}_{2,0}^{(S^{n+1})} \hookrightarrow \mathbb{Z}_{4,0}^{(S^{n+1})}$, which we achieve by “dividing by 2”. We get from this the formula

$$\text{Sq}^1\left(\sum_{i=1}^m x_i\right) = \sum_{1 \leq i < j \leq m} [c(x_i, x_j), d(x_i, x_j)] + [c(x_i, x_j), 1 - d(x_i, x_j)]$$

which concludes our construction.

We will conclude this section by noting the this construction generalizes to give a representation for the Bockstein homomorphism of the sequence

$$0 \rightarrow \mathbb{Z}_p \rightarrow \mathbb{Z}_{p^2} \rightarrow \mathbb{Z}_p \rightarrow 0$$

for any prime p . We have so far constructed it for $p = 2$, as we believe it to be the most representative and easiest to describe and understand. We will now describe how to change the construction to make it work for all p :

We define the space $L_{n,p} := \{(x_1, \dots, x_p) \in (S^n)^p \mid \sum_{i=1}^p x_i \neq 0\}$, where we again treat S^n as a subspace of $\mathbb{R}^{n+1}$. We can then define $c_p : L_{n,p} \rightarrow S^n$ as sending $(x_1, \dots, x_p)$ to $\frac{\sum_{i=1}^p x_i}{|\sum_{i=1}^p x_i|}$. We define $d_p : (S^n)^p \rightarrow I$ as

$$d_p(x_1, \dots, x_p) := \begin{cases} \max_{1 \leq i, j \leq p} (|x_i - x_j|) & \text{if } \max_{1 \leq i, j \leq p} (|x_i - x_j|) \leq \frac{1}{2} \\ \frac{1}{2} & \text{otherwise.} \end{cases}$$

From this, we can define $J_p : (S^n)^p \times I \rightarrow \mathbb{Z}_{p^2,0}^{(S^n \times I)}$ such that for all $x \in (S^n)^p$ and $t \in I$, we have

$$J_p(x, t) := \begin{cases} p(p-1)(c(x), td(x)) - p(p-1)(c(x), t(1-d(x))) & \text{if } x \in L_{n,p} \\ 0 & \text{otherwise.} \end{cases}$$

We can again use the pasting lemma to show J_p is continuous. We can then define $f' : \mathbb{Z}_{p,0}^{(S^n)} \times I \rightarrow \mathbb{Z}_{p^2,0}^{(S^n \times I)}$ for all $\sum_{i=1}^m x_i \in \mathbb{Z}_{p,0}^{(S^n)}$ and $t \in I$ as

$$f'(\sum_{i=1}^m x_i, t) := \sum_{i=1}^m (x_i, 0) - (x_i, t) + \sum_{1 \leq i_1 < i_2 < \dots < i_p \leq m} J(x_{i_1}, \dots, x_{i_p}, t).$$

We can prove that f' is well defined and continuous analogously to how we proved it for the $p = 2$ case. We can continue the construction analogously (to the $p = 2$ case), and in the end we will get a map $B_p : \mathbb{Z}_{p,0}^{(S^n)} \rightarrow \mathbb{Z}_{p,0}^{(S^n \times I / \sim)} \cong \mathbb{Z}_{p,0}^{(S^{n+1})}$ such that for all $\sum_{i=1}^m x_i \in \mathbb{Z}_{p,0}^{(S^n)}$, we have that

$$B_p(\sum_{i=1}^m x_i) = \sum_{1 \leq i_1 < \dots < i_p \leq m} (p-1)(c(x_{i_1}, \dots, x_{i_p}), d(x_{i_1}, \dots, x_{i_p})) - (p-1)(c(x_{i_1}, \dots, x_{i_p}), 1 - d(x_{i_1}, \dots, x_{i_p})).$$

The map B_p then represents the Bockstein homomorphism $H^n(X; \mathbb{Z}_p) \rightarrow H^{n+1}(X; \mathbb{Z}_p)$.

6 Conclusion

In this thesis we have constructed Configuration Homology, shown that it is a homology theory, and related it to several other homology theories: namely,

singular homology, the homology theory given by the Dold-Thom Theorem, and the homology theory induced by the spectrum $G_0^{(\Sigma^\infty S^0)}$. We used this same spectrum to define configuration cohomology, which we then used to represent several cohomology operations. These include the cup product and the Bockstein Homomorphism for the short exact sequence $0 \rightarrow \mathbb{Z}_2 \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2 \rightarrow 0$, which happens to be the Steenrod square Sq^1 . We then generalized this construction to work for the Bockstein Homomorphisms associated to $0 \rightarrow \mathbb{Z}_p \rightarrow \mathbb{Z}_{p^2} \rightarrow \mathbb{Z}_p \rightarrow 0$ for all primes p .

During the thesis, we introduced the tool of decomposition structures on topological groups. We believe this tool, alongside Theorem 2.12, to be useful for studying the homotopy-theoretic properties of topological groups (especially those groups that fall outside the scope of Lie theory) even outside of this thesis.

We would like to conclude by mentioning potential avenues for further exploring this topic. These avenues will involve loosening the conditions on the definition of the formal sum space $G^{(X)}$.

First, we will explore what happens when we do not require G to be discrete, just an arbitrary topological abelian group. The definition of a formal sum space, as well as the definition of configuration homology (including the long exact sequence in homology) naturally extends to this setting. However, our proof that configuration homology is a homology theory heavily relies on the coefficient group G being discrete. It is therefore interesting to ask: For what class of topological abelian groups is $H_*^{\text{conf}}(X; G)$ a homology theory?

There are two reasons the proof we presented does not generalize to non-discrete groups: First, our proof relies on a comparison to singular homology, which uses discrete groups. This can potentially be circumvented: we believe that Theorem 2.12 is potentially strong enough on its own to prove excision, without relying on singular homology (which is the only Eilenberg-Steenrod axiom that is difficult to show for configuration homology). The second problem is that Theorem 2.12 itself relies on the assumption that $G^{(X)}$ is locally decomposably contractible with respect to its defining decomposition structure $G(X) \rightarrow G^{(X)}$, which we proved using the assumption that G is discrete (and X is hyperlocally contractible). We believe that this proof can likely be altered to work in a more general case if we assume G to be locally contractible. To summarize, we make the following two conjectures:

Conjecture 6.1. *For any topological abelian group G , if for all hyperlocally contractible spaces X we have that $G^{(X)}$ is locally decomposably contractible with respect to its defining decomposition structure, then $H_*^{\text{conf}}(X; G)$ is a homology theory on CW-complexes.*

Conjecture 6.2. *If G is a locally contractible topological abelian group and X is a hyperlocally contractible space, then $G^{(X)}$ is locally decomposably contractible with respect to its defining decomposition structure.*

A proof of these conjectures would greatly generalize the results of this thesis.

A second way we can loosen the assumptions on $G^{(X)}$ is by not requiring G to be abelian. Unlike the previous generalization, the definition of a formal sum

space does not naturally extend to non-abelian groups. We can instead define the *free product space* $F(X; G)$ (where X is a space and G is a topological group) as a group with presentation

$$\langle G \times X \mid \forall g_1, g_2 \in G, x \in X, (g_1, x)(g_2, x) = (g_1 g_2, x), (e, x) = e \rangle$$

where e is the identity. We can define the topology on $F(X; G)$ analogously to how we defined the topology on the formal sum space. We can then define a series of functors $X \mapsto \pi_n(F(X; G))$, which we can think of as topological invariants associated to X . We believe that investigating this invariant, as well as how it relates to configuration homology, could be a fruitful line of research.

Finally, the third way we can loosen the definition of $G^{(X)}$ is by allowing the formal sums to be “twisted”. To be precise: Let X be a space and let G be a topological abelian group. Let $p : E \rightarrow X$ be a fiber-bundle over X with fiber G and structure group $\text{Aut}(G)$. Then we can define the *twisted presum space* to be $G_p(X) := \coprod_{n=1}^{\infty} E^n$. We can then define the *twisted formal sum space* $G_p^{(X)}$ to be $G_p(X) / \sim$, where $\sim$ is an equivalence relation we will now construct: There is a natural fiber bundle $G_p(X) \rightarrow \coprod_{n=1}^{\infty} X^n$ with fiber $\coprod_{n=1}^{\infty} G^n$. By local triviality, this will locally look like an open subset of $\coprod_{n=1}^{\infty} (G \times X)^n = G(X)$. On each of those subsets, two elements are related by $\sim$ if they represent the same formal sum in $G^{(X)}$, and this generates the relation $\sim$.

If G is discrete, we believe that $G_p^{(X)}$ is related to homology with local coefficients (the interested reader may find out about them in [5, p. 95-110]). Specifically, we conjecture the following:

Conjecture 6.3. *Let G be a discrete group, let X be a topological space, and let $p : E \rightarrow X$ be a fiber bundle with fiber G and structure group $\text{Aut}(G)$. Then $\pi_n(G_p^{(X)})$ is naturally isomorphic to the n th homology group of X with local coefficient p .*

We would like to also mention how this can potentially be used to analyse the homology of a fiber bundle. Given a fiber bundle

$$F \hookrightarrow E \rightarrow X$$

and a topological abelian group G , we believe that we can attain a short exact sequence of groups

$$(G_0^{(F)})_p^{(X)} \hookrightarrow G^{(E)} \rightarrow G^{(X)}$$

where $p : T \rightarrow X$ is a fiber bundle with fiber $G_0^{(F)}$. Intuitively, this is for the following reason: the kernel of $G^{(E)} \rightarrow G^{(X)}$ is a formal sum of points in $G^{(E)}$. If we restrict this sum to a fiber in E , it needs to add up to zero, and each such fiber is indexed by an element of X . Therefore, at least intuitively, every formal sum in the kernel of $G^{(E)} \rightarrow G^{(X)}$ can be expressed as a formal sum of points in X with coefficients in $G_0^{(F)}$, up to a twist induced by the fiber bundle $E \rightarrow X$. If this is true, and the sequence $(G_0^{(F)})_p^{(X)} \hookrightarrow G^{(E)} \rightarrow G^{(X)}$ induces a long exact

sequence in homotopy, then it could be possible to analyse the homology of the fiber bundle $F \hookrightarrow E \twoheadrightarrow X$ by examining the homotopy groups of $(G_0^{(F)})_p^{(X)}$.

We also believe that configuration cohomology is worth exploring in more detail. We note that many of the generalizations we have just mentioned also work for configuration cohomology.

We have, in this thesis, represented several cohomology operations using configuration cohomology. We think that representing all (or as many as possible) Bockstein Homomorphisms and Steenrod Squares (or powers) could provide further insight into their behaviour.

We hope to see, and contribute to, future research exploring this topic.

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